

Fair Property Tax: Targeting Regressivity in Machine Learning-Based Valuation

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Abstract

Property value assessments determine how local property-tax burdens are distributed among home owners. Assessor offices increasingly use machine learning to estimate values for large and heterogeneous property inventories, given their strong predictive power. However, predictive accuracy alone does not secure fairness in the estimated values. A common problem is known by assessors as *regressivity*: valuation-to-sale ratios can be systematically higher for lower-valued properties and lower for higher-valued properties. Working in collaboration with assessors at the Cook County Assessor’s Office (CCAO), we study whether this application-specific pattern can be reduced during model-training. We use the first-order association between log valuation errors and log sale prices as a regularization mechanism. We consider a direct squared-covariance penalty and also develop a sample-additive upper-bound surrogate that is compatible with standard boosting implementations. The evaluation of the corrected models is done using chronological validation, held-out and forward evaluation, assessor-facing ratio-study measures, mechanism and residual-structure diagnostics, and value-group ratio profiles. Across substantial portions of the square-covariance and Surrogate paths, increasing regularization attenuates the targeted first-order pattern and moves several assessor-facing vertical-equity diagnostics toward neutrality, while stronger regularization exposes tradeoffs in predictive performance and valuation ratio fairness. Near-neutral first-order or grouped diagnostics do not imply a flat value-related profile, and broader residual-price dependence can remain. A fixed-prediction-space analysis clarifies that the two objectives are related but not equivalent: the Direct penalty follows a rank-one covariance-correction path, whereas the Surrogate is a log-price-distance-weighted projection that can combine multiple learnable correction directions. Exploratory multi-jurisdiction analyses separately examine transferability and heterogeneity. We report complete regularization paths and do not select a deployment penalty strength.

Machine-learning mass-appraisal models can be predictively strong yet leave a systematic price-related trend in valuation ratios: the pattern assessors call *regressivity*. We study a narrow intervention in a research translation of the Cook County Assessor’s Office residential LightGBM workflow. The target is one global, first-order quantity: the covariance between log valuation errors and log sale price among observed sales, which is proportional to the classical log-ratio/log-price slope. A Direct penalty acts on that squared covariance; a Jensen-derived Surrogate is observation-additive and so fits the standard boosting custom-objective interface. In a fixed prediction space with an intercept, Direct is exactly a one-parameter centered spread map and the Surrogate a generally multi-directional weighted projection. We trace prespecified penalty grids with seven chronological rolling-origin folds, a later held-out block, and a separate 2025 forward period, holding the base learner fixed. A centered-spread post-hoc rescaling solved on development predictions reproduces much of Direct’s first-order movement over the range Direct attains, so training-time regularization is not necessary to move this trend; yet at matched first-order correction the families are not interchangeable in residual structure, and no family is uniformly better across the reported diagnostics. Across those folds, the held-out block, and 2025 the correction-strength relationship does not transport, so no penalty strength is selected. The contribution is one auditable first-order mechanism for observed sales with its temporal limits, not a fairness guarantee or a claim about tax burdens.

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1 Introduction

Property taxes help pay for public schools, roads, and other local services in the United States [29, 5]. The tax system relies in part on estimates of property value. If some properties are valued systematically too high and others too low, the resulting assessments can shift tax burdens among owners [5, 29].

Assessors use *mass appraisal* to value large groups of properties as of a common date, using shared data, methods, and statistical checks [31]. The statistical models used for this purpose are often called automated valuation models (AVMs). Recent research studies machine-learning methods, including gradient-boosted trees, alongside traditional regression models because they can predict sale prices well [25, 46, 48, 57]. Prediction, however, is only part of the assessment task. Assessors also need to know whether valuations are consistent across properties and across the price range.

Ratio studies provide one way to check this. They compare valuations with observed sale prices, which serve as proxies for market value [27, 31]. The model studied here estimates *fair market value* before that value is converted to an assessed value and used in later steps that determine a tax bill [15, 17, 16]. We therefore use the *valuation-to-sale ratio*: the model’s estimated market value divided by the observed sale price. We call this the *valuation ratio* below. Ratio studies examine both variation among comparable properties (horizontal uniformity) and whether ratios change with property value (vertical equity) [27, 31, 32].

Our focus is *regressivity*, a form of vertical inequity in which lower-valued properties tend to have higher valuation ratios than higher-valued properties. Such a pattern can place a greater relative assessment burden, and potentially a greater tax burden, on owners of lower-valued properties [40, 5, 3]. Because a property’s market value is not directly observed, we study this pattern among sold properties using sale price as a value proxy [32]. A good overall prediction score cannot show whether valuation ratios vary systematically across the value range.

This paper grew out of a collaboration with the Cook County Assessor’s Office (CCAO), whose residential valuation workflow uses a county-wide LightGBM model trained on screened sales [15, 33]. The model uses property, location, and market information to estimate fair market value. Assessors must value a much wider inventory than the properties that sell, and the available sales must be checked before they can be used for model training and evaluation. We study a Python research translation of the office’s published workflow and evaluate it on verified sales. These are research results for the valuation stage, not official assessments, estimates of final tax bills, or a deployment recommendation. Any useful correction must also fit the sales-based modeling, temporal validation, and review processes in which assessors work [30, 28, 31].

The motivation is visible within the current model class. On the primary held-out sales, ordinary LightGBM achieves a price-scale R_P^2 of 0.894, yet its price-related bias (PRB) is -0.091 , outside the conventionally preferred $[-0.05, 0.05]$ range (Section 2.4). PRB measures how valuation ratios vary with a combined proxy for property value, based on sale price and the model’s valuation. A negative PRB indicates that ratios tend to decline as this value proxy increases. We ask whether this pattern can be reduced while retaining predictive performance. The earlier linear-regression workflow provides historical context; this paper evaluates corrections within the LightGBM model class.

For training, we use an established *first-order* summary: the overall straight-line slope of log valuation ratios against log sale prices [11]. A negative slope means that ratios tend to fall as sale prices rise. This training slope is proportional to the covariance between log-price prediction errors and log sale prices; PRB is a separate statistic based on a combined value proxy. Sale price appears both in the prediction error and in the ratio. Even a well-fitted model can therefore have a negative error–sale-price association when individual sale prices contain unpredictable variation. The slope is a training target, not proof of inequity in unobserved market values, and a zero slope is not a guarantee of fairness. We assess the resulting valuations with separate out-of-time ratio and residual-pattern diagnostics.

We study two ways to use that training target. The *Direct* objective penalizes the squared covariance across the sales used for fitting. The *Surrogate* uses an upper bound that turns the penalty into weighted squared error: an error receives more weight when a sale’s log price is farther from the training average. This loss separates across sales, making it compatible with standard boosting software. The objectives can therefore change predictions in different ways. Section 3 gives the derivation and the fixed-prediction-space analysis of that difference.

The ratio–price diagnostic and dependence penalties build on earlier assessment and fair-regression work,

including related corrections outside property valuation [11, 34, 53]. We retain the same sales, predictor information, prediction target, and base-learner configuration across each penalty path. Ordinary LightGBM is the workflow benchmark. The Direct and Surrogate paths also share an unpenalized custom-objective fit, which is the reference for measuring penalty effects: it differs numerically from ordinary LightGBM even though both are unpenalized.

We trace both training paths through chronological validation, a later held-out sample, and a separate 2025 forward sample after refitting on eligible 2016–2024 sales. We also compare them with a simple post-hoc rescaling of the shared unpenalized custom-objective predictions, which does not require retraining. We match the three approaches on the first-order slope measured in development validation, then compare prediction, assessor-facing equity measures, and remaining ratio patterns on later sales. Matching during development does not make their later slopes equal. Separate exploratory analyses examine variation across other jurisdictions. We do not select a penalty strength for deployment.

In this application, portions of both training paths reduce the targeted trend without lowering price-scale R_P^2 relative to their shared unpenalized fit. Stronger correction can lower accuracy, while other patterns across the value range can remain. Post-hoc rescaling reproduces much of Direct’s first-order change over the range Direct attains. At similar development slopes, however, the approaches can leave different residual patterns, and none performs best on every measure. The paths also change across evaluation periods: the same penalty strength need not have the same effect later. These findings concern an overall price-related pattern among observed sales. Neither objective, by itself, sets the overall valuation level, removes nonlinear or local inequity, or establishes changes in final tax liabilities.

The paper’s contribution is a CCAO-centered evaluation of how to reduce a price-related valuation-ratio pattern within an existing model class. In a research translation of the office’s residential LightGBM workflow, we turn a classical first-order diagnostic into two training objectives, explain why they can produce different predictions, and compare both with post-hoc rescaling at matched development slopes. We then examine the resulting accuracy and equity paths on later sales, including their temporal limits. This comparison shows what retraining adds and where it does not, without selecting a penalty strength for deployment.

The remainder of this section reviews related work. Section 2 describes the CCAO setting and evaluation measures; Section 3 develops the objectives. The subsequent sections present the empirical design, results, and practical implications.

1.1 Related Work

Machine Learning in Mass Appraisal Recent reviews document the growing research on automated valuation models and machine learning (ML) in mass appraisal and real estate valuation. Multiple regression remains common, alongside tree ensembles such as random forests and gradient boosting, as well as neural-network-based methods [54, 46, 57, 2]. These reviews organize the work they survey primarily by predictive criteria, such as out-of-sample accuracy and performance relative to alternative algorithms. Vertical equity, although routinely evaluated in assessor practice, is not among the criteria around which they organize model training or model selection [54, 46, 57]. Tree ensembles are among the methods these reviews both survey and record as predictively competitive [46, 48, 57]. A practical training-time equity intervention should therefore be compatible with the model classes these reviews document rather than require a specialized valuation architecture.

Operational adoption by assessment offices nevertheless remains partial and uneven. Bidanset and Rakow [6] identify communication challenges, limited technical capacity, and budget constraints among perceived barriers to wider adoption. Moreover, predictive modeling is only one component of the assessment workflow; the use of ML does not replace sales verification, out-of-time evaluation, ratio studies, model interpretation, or administrative review [30, 25]. The IAAO Artificial Intelligence Task Force [25] emphasizes explainability, defensibility, administrative feasibility, and integration into existing workflows as central conditions for operational use of advanced ML models. These considerations favor equity-aware methods that can be incorporated into existing model-training, validation, and review pipelines with minimal disruption.

Vertical equity and regressivity. Equity has long been a central objective in mass appraisal practice and standards. The assessor literature distinguishes *horizontal equity*, which concerns the uniform treatment of

similar properties, from *vertical equity*, which concerns systematic differences in assessment outcomes across property-value levels [27, 18, 41]. A common form of vertical inequity is *regressivity*: assessment-to-sale ratios tend to decline with property value, so lower-valued properties tend to be relatively overassessed and higher-valued properties tend to be relatively underassessed. Ratio studies evaluate assessment uniformity using complementary diagnostics. For example, the Coefficient of Dispersion (COD) summarizes dispersion in assessment ratios, whereas the Price-Related Differential (PRD) and Price-Related Bias (PRB) are commonly used to diagnose price-related vertical inequity [27, 26]. The statistical measurement of price-related vertical inequity has a long history in this literature. Paglin and Fogarty [42] reframed property-tax equity around the distribution of assessment-to-value ratios; Cheng [11] asked how assessment performance itself should be measured and, in doing so, supplied the direct antecedent of the diagnostic this paper trains on: he writes a log assessment-to-sale ratio as a function that includes the log settlement price, entering at coefficient $\theta - 1$, so a log-ratio/log-price slope of exactly this form is already present in the classical assessment literature. Cheng also discusses how appraisal lag and heterogeneous value growth can affect the observed ratio-price relationship. Edelstein [20] appraised residential property-tax regressivity directly; and Sunderman et al. [51] developed statistical tests for vertical inequity. That work supplies both the ratio-based framing used here and the recurring lesson that alternative specifications of the same test need not agree.

No single diagnostic fully characterizes vertical equity. Methodological studies show that vertical-equity measures can differ materially in their statistical behavior, and that some regression-based diagnostics can be biased when assessed values are themselves generated by regression-based valuation models [26, 41]. Accordingly, the assessor literature recommends examining multiple complementary diagnostics rather than relying on a single global measure. The classical ratio-price slope is therefore one member of a complementary set that also contains the price-related differential, price-related bias, the modified Kakwani index and the proposed exposure-draft vertical-equity index; this paper trains on the slope and continues to report the others, rather than treating any one of them as a complete account of vertical equity. Empirical studies further document substantial regressivity and consequential distributional disparities across jurisdictions [5, 3]. Cook County is also the setting for direct evidence on algorithmic assessment reform: Liu [38] exploits the staggered triennial reassessment cycle around the office’s adoption of a county-wide machine-learning valuation model and reports a substantial flattening of the log assessed-to-sale-price gradient afterwards. That is external evidence that reforming an assessment algorithm can materially change measured price regressivity; it evaluates none of the objectives studied here and supplies no benchmark for them. Recent large-scale evidence also shows that predictive accuracy and assessment equity need not be intrinsically opposed: improvements in the information available to valuation models can, in many settings, improve both simultaneously [49].

A smaller literature moves beyond aggregate diagnosis toward model specification, localized validation, and explicit correction intended to improve assessment equity. Proposed approaches include spatial and geographically weighted methods, stratified procedures, ratio-study-based model validation, and fairness-aware valuation models [44, 7, 8, 23, 24, 10]. These studies approach assessment equity along several distinct dimensions: the modeling architecture, the information set, the spatial or stratification structure, the validation design, or an explicit equity or fairness formulation written into the estimation problem. The route taken here is narrower and different rather than better: one first-order training target inside an existing learner, with no change to architecture, features, or sample construction. No experiment in this paper combines it with any of those alternatives, so no claim of compatibility with them is made. A closely related approach is Candogan, Han, and Lu [10], who develop a *K-segment* property-valuation model that explicitly targets regressivity and studies the associated accuracy-fairness relationship. Our approach addresses the same assessor-side objective through a different mechanism: rather than introducing a segmented valuation architecture, we use a first-order log-error and log-sale-price association as a training target within an existing regression learner. The method therefore targets a deliberately narrow first-order mechanism during training while retaining standard ratio studies, proxy-aware vertical-equity diagnostics, and subgroup review for out-of-sample evaluation.

Fair regression and dependence-based constrained learning. Although much of the fairness-in-ML literature initially focused on classification, *fair regression* studies how application-specific fairness requirements can be incorporated when predictions are continuous, as in property valuation. Existing approaches consider criteria such as statistical parity, bounded group loss, mean parity, or distributional parity and

implement them through constrained or regularized empirical risk minimization, post-processing, or recalibration [1, 21, 13, 12, 56]. The broader methodological principle is that predictive risk can be optimized subject to, or jointly with, an application-specific fairness criterion. Varying the strength of this requirement produces a family of solutions that makes the relationship between predictive performance and the targeted fairness criterion explicit, without requiring that improvements in one necessarily worsen the other.

A particularly relevant strand expresses fairness requirements as restrictions on statistical dependence between predictions and a sensitive variable. Prior work has quantified or controlled such dependence using correlation and coefficients of determination [34], kernel-based measures such as the Hilbert–Schmidt Independence Criterion (HSIC) [43, 37], and maximal-correlation criteria [39, 36]. Related regularization approaches also control the contribution of sensitive attributes to regression predictions [47]. These approaches are particularly relevant when the variable of concern is continuous and can often be incorporated directly into regression objectives. Our application, however, differs from conventional group-fairness settings. Sale price is the observed transaction-based proxy for market value and the prediction target, not a protected attribute, and our objective is not demographic parity. We adapt the broader dependence-control principle to a narrower structural target: the signed first-order association between log valuation errors and log sale prices. The assessor literature supplies the domain-specific objective—reducing value-related regressivity—while the fair-regression literature supplies a general mechanism for incorporating statistical associations into model training.

A global slope correction is also not the only way to address regressivity: segmented valuation, richer spatial and temporal information, and post-hoc affine recalibration [4] are close alternatives, and the last of these is close enough that it is carried through this paper as an explicit comparator rather than left as a possibility. The narrower methodological contribution is to use that classical first-order relationship as a trainable target in the CCAO workflow, derive and distinguish the Direct and observation-additive Surrogate objectives, and evaluate their temporal paths and failure modes. Because a centered post-hoc spread map is the exact fixed-space form of the Direct correction, the comparison against it is reported here rather than deferred, and no claim that in-processing is preferable is made on the strength of it.

Residual–outcome decorrelation outside mass appraisal. The mechanism controlled here, namely dependence between a model’s residuals and the very outcome that model predicts, has already been formulated, and controlled, outside mass appraisal. Some of the closest methodological prior art comes from outside the valuation literature, particularly brain-age prediction, where the same structure appears as the “brain-age delta” problem: the residual of predicted age on true age is systematically associated with true age. Smith et al. [50] set out that estimation problem together with its regression-based correction, and Beheshti et al. [4] give a post-hoc affine bias-adjustment scheme for it. Treder et al. [53] impose a correlation constraint directly on the training objective and trace a tunable family of solutions as the constraint strength varies. Wang et al. [55] instead reshape the loss so that it corrects the target-dependent component of predictive bias, and Ren et al. [45] place an explicit delta–outcome correlation loss inside a nonlinear learner. Lee and Chen [35] supply a statistical framing for this class of procedures, through outcome-calibrated regression and inference based on predicted outcomes. Related fair-regression work also expresses dependence restrictions through measures such as the coefficient of determination [34].

The novelty boundary. The log-ratio/log-price diagnostic is established in assessment research [11]. Dependence penalties, target-dependent corrective losses, and post-hoc affine corrections also have precedents in regression [34, 53, 50, 55, 45, 4, 35]. Our contribution is the CCAO-centered evaluation of a narrow correction within the existing LightGBM model class. We derive two training objectives from the classical diagnostic, explain their different fixed-prediction-space mechanisms, compare their out-of-time paths with post-hoc rescaling at matched development slopes, and report where the results vary across periods. This study concerns observed sales in a research translation of the published workflow; it does not select a penalty strength for deployment.

2 Preliminaries: The CCAO Baseline and Empirical Motivation

We describe the CCAO valuation setting, define the measures used to evaluate the models, and show the baseline result that motivates the correction.

2.1 The CCAO Residential Valuation Workflow

The CCAO values a large and heterogeneous residential inventory on a triennial reassessment cycle [14]. Its residential modeling evolved from township-level linear-regression models, with linear-regression and gradient-boosting alternatives used in 2019–2020, to a county-wide LightGBM model beginning in 2021 [15]. LightGBM is an implementation of gradient-boosted decision trees [33]. That transition has itself been studied as an algorithmic assessment reform: using the staggered triennial reassessment cycle and the log assessed-to-sale-price ratio, Liu [38] reports a substantial flattening of the price gradient afterwards.

The CCAO’s current published residential automated valuation model (AVM) uses verified sales and a broad set of property, spatial, neighborhood, market, and contextual predictors to estimate residential market values [15]. The prediction model is embedded in a broader operational workflow comprising data preparation, model training, valuation, evaluation, interpretation, and final review. Candidate valuations are evaluated using both predictive-performance measures and assessor-specific ratio-study statistics across multiple geographic areas and property classes.

Our analysis compares ordinary LightGBM with corrections to the same model class. We do not re-estimate the earlier linear-regression workflow or reconstruct official historical assessments. The models compared here use the same prediction target, underlying feature information, development sample, temporal design, and evaluation procedures. Section 4.1 gives the sample construction and validation details.

2.2 Prediction Setting and the Baseline Model

Let $\mathcal{I}$ denote the set of residential sales retained after the CCAO sales-validation and transaction-eligibility procedures described in Section 4.1. For each $i \in \mathcal{I}$, we observe

$$(x_i, P_i, t_i),$$

where x_i contains property and parcel characteristics; tract-level socioeconomic and demographic variables from the U.S. Census Bureau’s American Community Survey (ACS); and geographic, neighborhood, environmental, accessibility, market, and temporal features.

Here $P_i > 0$ is the verified sale price, our observed proxy for the property’s unobserved market value, and t_i is the sale date used to construct the chronological samples [15, 32]. Any time-derived predictors are included in x_i .

Since residential sale prices are positive and highly right-skewed, the baseline model is trained on the log-price scale. Define

$$y_i = \log P_i, \quad \hat{y}_i = f(x_i), \quad e_i(f) = \hat{y}_i - y_i, \quad (1)$$

where $f \in \mathcal{H}$ is a candidate prediction function and $e_i(f)$ is the *log valuation error*, defined with prediction minus observed log price so that it coincides with the log valuation ratio below. The corresponding prediction on the original price scale is

$$\hat{P}_i = \exp(f(x_i)). \quad (2)$$

We evaluate predictions on the price scale using $\hat{P}_i$ from (2). Direct exponentiation is an operational retransformation, not a correction for conditional-mean bias. Define the *valuation-to-sale ratio* as

$$r_i(f) = \frac{\hat{P}_i}{P_i} = \exp(e_i(f)). \quad (3)$$

We call $r_i(f)$ the *valuation ratio*. Its numerator is the model-estimated market value, before any downstream statutory assessment adjustments. By construction, $e_i(f) = \log r_i(f)$: positive values mean the predicted value exceeds the sale price, and negative values mean it falls below it. For a fixed model we write r_i and e_i .

The predictive baseline is ordinary LightGBM, a gradient-boosted tree model representing CCAO’s current model class. For a training sample of size n , it minimizes mean squared log-price error:

$$L(f) = \frac{1}{n} \sum_{i=1}^n e_i(f)^2. \quad (4)$$

This loss is the predictive component of the objective modified in Section 3. LightGBM fits regression trees sequentially and can represent nonlinearities and interactions [15, 33]. The ordinary loss has no term that directly controls value-related trends in valuation ratios. Section 2.4 shows why such a term is worth studying.

2.3 Predictive, Assessor-Facing, and Residual-Structure Diagnostics

For an evaluation sample $S \subseteq \mathcal{I}$, write $n_S = |S|$. We assess prediction on both the log-price training scale and the original price scale. Ratio-study measures use original-scale valuations and answer a different question from predictive accuracy.

Predictive performance. We report mean absolute error (MAE) in dollars, mean absolute percentage error (MAPE) relative to sale price, and R_P^2 on the original price scale. Root mean squared error on log price, $\text{RMSE}_{\log P}$, measures error on the training scale. MAPE compares ratios with one; COD below measures their dispersion around the sample median. Exact formulas are in Appendix A.

Valuation level. Recall from (3) that $r_i = \hat{P}_i/P_i$. A ratio above one indicates that the predicted value exceeds the observed sale price, whereas a ratio below one indicates that it falls below the sale price. Define the median, arithmetic mean, and price-weighted mean ratios as

$$m_S = \text{median}_{i \in S}\{r_i\}, \quad \bar{r}_S = \frac{1}{n_S} \sum_{i \in S} r_i, \quad \bar{r}_{W,S} = \frac{\sum_{i \in S} \hat{P}_i}{\sum_{i \in S} P_i} = \frac{\sum_{i \in S} r_i P_i}{\sum_{i \in S} P_i}. \quad (5)$$

These ratios describe aggregate valuation level, with one as the neutral value. The median is less sensitive to extremes; the arithmetic mean weights sales equally; and the weighted mean gives more weight to higher-price sales. Agreement on overall level does not establish uniformity or vertical equity.

Ratio dispersion (Horizontal uniformity). Ratio dispersion is an important assessor-facing dimension of horizontal uniformity. We summarize it using the coefficient of dispersion (COD) and coefficient of variation (COV). The COD is

$$\text{COD}(S) = 100 \times \frac{1}{n_S} \sum_{i \in S} \frac{|r_i - m_S|}{m_S}, \quad (6)$$

The coefficient of variation (COV) measures the standard deviation of ratios relative to their mean; its formula is in Appendix A. Lower values indicate less aggregate ratio dispersion, but reference ranges depend on property class and market conditions, and unusually low dispersion can itself warrant review [27, 32]. To avoid ambiguity with the covariance penalty introduced later, COV always denotes the coefficient of variation, whereas statistical covariance is written as $\text{Cov}(\cdot, \cdot)$.

Regressivity (Vertical equity). Vertical-equity measures ask whether valuation ratios change across the value range. True market value is unobserved, so each measure relies on a sale-based value proxy and can give a different answer [26, 32]. We report PRD, PRB, MKI, and VEI together. Table 1 distinguishes adopted reference ranges from the proposed May 2026 exposure-draft ranges; none is treated as a finding of legal compliance.

The *price-related differential* (PRD) is

$$\text{PRD}(S) = \frac{\bar{r}_S}{\bar{r}_{W,S}}. \quad (7)$$

Because the weighted mean gives more weight to higher-price sales, PRD above one suggests a regressive direction. The adopted reference range is $[0.98, 1.03]$ [27]. High-value sales and extreme ratios can affect PRD, so we read it alongside the other measures [32].

The following regression- and group-based vertical-equity diagnostics require a proxy for the property's value level. Ranking properties only by sale price can mechanically bias ratio-based vertical-equity tests toward finding regressivity because sale price also appears in the denominator of r_i , while using valuation alone can induce bias in the opposite direction. Following IAAO guidance, we therefore use the equal-weight market-value proxy

$$V_i^{\text{proxy}} = \frac{1}{2} \left(P_i + \frac{\hat{P}_i}{m_S} \right), \quad (8)$$

which combines the verified sale price with the median-normalized predicted value to reduce this source of proxy-induced bias [27, 32].

Both PRB and VEI use the equal-weight proxy in Eq. (8). Appendix A records the exposure draft's notation discrepancy and the check against the executed metric code.

We first report the *price-related bias* (PRB). Let $\tilde{V}_S^{\text{proxy}}$ denote the median of V_i^{proxy} over S . The PRB is the slope in the simple linear regression

$$\frac{r_i - m_S}{m_S} = a_S + \text{PRB}(S) \cdot \log_2 \left(\frac{V_i^{\text{proxy}}}{\tilde{V}_S^{\text{proxy}}} \right) + \varepsilon_i. \quad (9)$$

Negative PRB values indicate that valuation ratios tend to decline as the market-value proxy increases and are therefore associated with regressivity. Values between -0.05 and 0.05 are conventionally preferred [27].

The *Modified Kakwani Index* (MKI) compares the concentration of predicted values, ordered by sale price, with the distribution of sale prices. When the index is defined, a value of one indicates neutrality on this measure; values below one point in the regressive direction [32]. Its sale-price ordering can affect the result when sale price imperfectly measures market value. We therefore read MKI alongside the other measures; its construction is in Appendix A.

Finally, we report the *Vertical Equity Indicator* (VEI), the quantile-grouped vertical-equity measure emphasized in the May 2026 IAAO exposure draft [32]. The draft varies the number of percentile groups with sample size: for the large evaluation samples used in the main countywide analysis ($n_S \geq 501$), observations are divided into deciles according to V_i^{proxy} . Let S_{q_1} and $S_{q_{10}}$ denote the lowest and highest deciles, and let $m_{S_{q_1}}$ and $m_{S_{q_{10}}}$ denote their median valuation ratios. The VEI is

$$\text{VEI}(S) = 100 \times \frac{m_{S_{q_{10}}} - m_{S_{q_1}}}{m_S}. \quad (10)$$

A negative VEI indicates that higher-value properties have lower median valuation ratios than lower-value properties. The exposure draft proposes a reference band of $[-10\%, 10\%]$, together with confidence-interval-based testing for point estimates outside that band.

Because a comparison of the first and last deciles can miss patterns between them, we also show every decile's median ratio and a descriptive 90% interval.

The following mechanism and residual-structure diagnostics have no assessor-standard reference ranges.

Mechanism and residual-structure diagnostics. These three diagnostics have distinct roles: $\beta_{\log}$ measures the targeted first-order trend, Δ_{NL} measures nonlinear conditional-mean structure beyond it, and distance correlation measures broader dependence.

First-order mechanism. We first report the slope from regressing log valuation ratios on log sale prices:

$$\beta_{\log}(S) = \frac{\widehat{\text{Cov}}_S(e, y)}{\widehat{\text{Var}}_S(y)} = \frac{\widehat{\text{Cov}}_S(\log r, \log P)}{\widehat{\text{Var}}_S(\log P)}. \quad (11)$$

A value of zero indicates no first-order log-ratio–log-sale-price trend. Negative values indicate the regressive direction and positive values the progressive direction. For a fixed evaluation sample, $\widehat{\text{Var}}_S(y)$ is common

Table 1: Predictive and assessor-facing evaluation measures.

Measure	Ideal value	Reference range	Guidance status	Interpretation
Predictive performance				
R_P^2	1	–	n/a	Original price-scale goodness of fit.
MAE_P	0	–	n/a	Mean absolute prediction error in dollars.
$MAPE_P$	0	–	n/a	Mean absolute prediction error relative to sale price.
$RMSE_{\log P}$	0	–	n/a	RMSE on the log-price estimation scale.
Valuation level				
Median ratio m_S	1	[0.90, 1.10]	IAAO 2013 (adopted)	Robust aggregate valuation level.
Mean ratio $\bar{r}_S$	1	[0.90, 1.10]	IAAO 2013 (adopted)	Unweighted aggregate valuation level.
Weighted mean $\bar{r}_{W,S}$	1	[0.90, 1.10]	IAAO 2013 (adopted)	Value-weighted valuation level.
Horizontal uniformity				
COD	–	[5.0, 15.0]	IAAO 2013 (adopted)	Absolute dispersion relative to the median ratio.
COV	–	[6.25%, 18.75%]	derived approximation	Relative dispersion around the mean ratio.
Vertical equity				
PRD	1	[0.98, 1.03]	IAAO 2013 (adopted)	Mean-to-weighted-mean ratio comparison.
PRB	0	[−0.05, 0.05]	IAAO 2013 (adopted)	Regression-based value-related ratio trend.
MKI	1	[0.95, 1.05]	May 2026 (exposure draft)	Distributional concentration relative to sale prices.
VEI	0	[−10%, 10%]	May 2026 (exposure draft)	High-versus-low value-group median-ratio difference.

Notes. “–” means that no universal ideal or range is imposed. The level-ratio, COD, PRD, and PRB ranges follow adopted IAAO guidance [27, 28, 31]. The COV band is a derived approximation from $COV \approx 1.25$ COD under approximately normal ratio distributions, not a published standard. The MKI band is CCAO implementation guidance also described as optimal for lower-variability strata in the May 2026 exposure draft; that draft proposes the VEI band. Neither is an adopted standard [32]. Appendix A gives further guidance details.

across models, so $\beta_{\log}(S)$ is the signed first-order slope corresponding directly to the covariance targeted during training. The slope predates this paper [11]. It is a training-mechanism diagnostic rather than a standalone measure of vertical equity: sale price is an imperfect value proxy and appears in the denominator of r_i .

Nonlinear conditional-mean structure. Let Z be standardized log sale price. The correlation ratio $\eta^2(e \mid Z)$ measures how much log-error variation is explained by its conditional mean given Z ; $\text{Corr}(e, Z)^2$ measures the part explained by the best straight-line relationship [19]. Their difference is the *correlation-ratio nonlinearity gap*:

$$\Delta_{NL} := \eta^2(e \mid Z) - \text{Corr}(e, Z)^2. \quad (12)$$

At the population level the gap is nonnegative and is zero when average log error has a straight-line relationship with log sale price. A smaller $|\beta_{\log}|$ therefore need not mean that curved patterns have disappeared. Appendix A gives the exact definitions and estimator.

Broader dependence. Distance correlation $d\text{Cor}(e, y)$ checks for broader dependence between log valuation errors and log sale prices [52]. Unlike $\beta_{\log}$, it has no regressive or progressive sign; it is a residual-structure diagnostic, not an assessor-facing equity criterion. Appendix A gives its identity and implementation.

The measures are descriptive diagnostics, not findings of standards compliance. We compute them on

chronological validation folds before aggregating the paths; countywide summaries can hide local differences. The training objectives do not optimize the full suite of evaluation measures.

2.4 Baseline Level and Empirical Motivation

On held-out sales, ordinary LightGBM attains $R_P^2 = 0.894$, but its valuation ratios decline across the value range: PRB is -0.091 and VEI is -26.5% . The pattern persists in the 2025 forward evaluation (PRB = -0.106 ; 95% confidence interval $[-0.111, -0.101]$). This motivates the question studied here: can we reduce the value-related ratio pattern while retaining the predictive strength of the current model class?

Table 2 reports the unpenalized LightGBM benchmark on the held-out sample and after refitting on all eligible 2016–2024 sales for the separate 2025 evaluation. The 2025 sample is not used for model selection; Section 4.1 gives the full design. Complementary valuation-level, ratio-dispersion, and residual-structure results appear in Appendix Table 10.

Table 2: Prediction and vertical-equity measures for ordinary LightGBM (standard raw-label native) on the held-out and 2025 forward evaluations. These are unpenalized benchmark results.

Measure	Held-out evaluation	2025 forward evaluation
Prediction		
R_P^2	0.894	0.904
MAE _P	\$75,655	\$78,484
MAPE _P	21.2%	20.8%
RMSE _{log P}	0.289	0.278
Vertical equity		
PRD	1.069	1.079
PRB	-0.091	-0.106
MKI	0.923	0.907
VEI	-26.5%	-28.6%

Notes. The benchmark is defined in Section 4.3. No linear-regression model is estimated or reported in this study. The guidance status and reference ranges are in Table 1; MKI and VEI use proposed exposure-draft guidance. These point estimates do not establish standards compliance or statistical significance. MAPE and VEI are in percent; MAE is in dollars. For the held-out evaluation the model is fit on the 344,607-sale development pool; for the 2025 forward evaluation it is refit on all 382,897 eligible 2016–2024 sales.

Figure 1 shows the ratio pattern across sale prices; the corresponding value-proxy decile profile appears in Appendix Figure 15. The baseline results describe the observed sample, not standards compliance or how a linear model would perform under this study’s design.

The penalty developed next targets squared covariance in Eq. (11). On a fixed training sample, log-sale-price variance is common to all fitted models, so reducing the magnitude of covariance reduces the magnitude of the corresponding first-order slope. PRB, VEI, and the other assessor-facing measures remain separate out-of-sample checks.

3 Covariance-Guided Regressivity Correction

This section translates the value-related regressivity identified in Section 2 into a training-time correction penalty. We first define a global first-order covariance-based regressivity target on the log-price scale, then introduce a direct squared-covariance penalty, and finally derive a sample-additive upper-bound surrogate that is compatible with standard boosting software. The correction penalty leaves the prediction target, feature information, and base model class unchanged. That is, for a fixed penalty formulation, the only additional tuning parameter is the penalty strength, which we denote ρ .

Let $J_0(f)$ denote the baseline training objective, whose data-fit component is the squared log-price loss $L(f)$ defined in Section 2.2. All quantities below are computed from the “current” training sample. In temporal validation, the centering quantities and penalties are recomputed using the training block of each fold only.

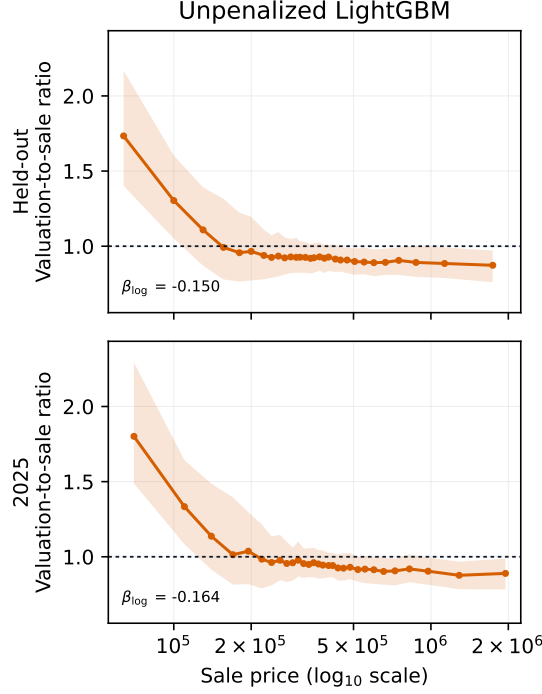


Figure 1: Descriptive valuation-ratio profiles against sale price for ordinary LightGBM in the held-out and 2025 forward evaluations. Curves show median ratios in 30 equal-count sale-price bins; shaded regions show interquartile ranges.

3.1 Log-Scale Regressivity Target

Let

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i, \quad c_i = y_i - \bar{y}, \quad \bar{e}(f) = \frac{1}{n} \sum_{i=1}^n e_i(f). \quad (13)$$

The empirical covariance between log-residuals and log-prices is

$$C(f) = \widehat{\text{Cov}}(e(f), y) = \frac{1}{n} \sum_{i=1}^n (e_i(f) - \bar{e}(f))(y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n e_i(f) c_i. \quad (14)$$

The last equality follows from $\sum_{i=1}^n c_i = 0$, so centering the residuals does not change the covariance.

This covariance has a direct slope interpretation. Let

$$V_y = \frac{1}{n} \sum_{i=1}^n c_i^2.$$

Provided $V_y > 0$, the least-squares slope from regressing log valuation ratios on log prices is

$$\text{slope}(e(f) \sim y) = \frac{C(f)}{V_y}. \quad (15)$$

Because V_y is fixed by the training sample, reducing $|C(f)|$ is equivalent to reducing the magnitude of this global linear trend. In particular, $C(f) < 0$ indicates that log valuation ratios tend to decline with log sale price, which is the first-order pattern associated with regressivity.

A qualification is essential because sale price is both the prediction target and the observed value benchmark. Let $m(X) = \mathbb{E}[Y \mid X]$ be the population squared-error predictor and let $e^* = m(X) - Y$. Then $\mathbb{E}[e^*] = 0$ and

$$\text{Cov}(e^*, Y) = -\mathbb{E}[\text{Var}(Y \mid X)] \leq 0. \quad (16)$$

Thus, even the Bayes predictor has negative residual–outcome covariance whenever sale prices contain conditionally unpredictable variation. Appendix D gives the finite-sample analogue: in a fixed linear prediction space containing an intercept, ordinary least squares satisfies $C_0 = -\|f_0 - y\|_n^2 \leq 0$. Driving training covariance toward zero can therefore spread predictions toward realized transaction prices rather than recover latent market values. Negative covariance is not, by itself, evidence of latent vertical inequity, and zero covariance is not a fairness condition. We use $C(f)$ only as a controllable first-order mechanism and judge the resulting valuations with chronological out-of-sample, proxy-aware ratio and residual-structure diagnostics.

This slope is a training-scale mechanism measure, not the assessor-facing PRB statistic defined in Section 2.3. We use covariance rather than correlation because V_y is fixed within the training sample, whereas correlation additionally normalizes by the model-dependent residual variance. Covariance therefore provides a direct training target for the global value-related slope.

Sale price enters this construction as the observed market-value benchmark for verified training sales; it is not introduced as a protected attribute. The broader connection to dependence-based fair regression is discussed in Section 1.1.

3.2 Direct Squared-Covariance Penalty

We first penalize the chosen first-order target directly. The *covariance-penalized learner* is

$$f_\rho^{\text{cov}} \in \arg \min_{f \in \mathcal{H}} \left\{ J_0(f) + \frac{\rho}{2} C(f)^2 \right\}, \quad \rho \geq 0. \quad (17)$$

When $\rho = 0$, formulation (17) reduces to the baseline learner. As ρ increases, the optimization places greater weight on reducing the global residual–price trend. Squaring the covariance treats negative and positive trends symmetrically. This is important because the objective is not to replace regressivity with progressivity, but to move the first-order relationship toward zero.

For interpretation, it is enough to consider the contribution of the covariance penalty to the gradient of observation i :

$$\frac{\partial}{\partial y_i} \left[\frac{\rho}{2} C(f)^2 \right] = \frac{\rho}{n} C(f) c_i. \quad (18)$$

Suppose, for example, that the fitted model is regressive due to $C(f) < 0$. For a sale below the geometric-mean price, $c_i < 0$, and thus the covariance component of the gradient is positive, i.e., a descent step lowers its predicted value. For a sale above the geometric-mean price, $c_i > 0$, and thus the covariance component is negative, i.e., a descent step raises its predicted value. The penalty therefore acts in the direction required to flatten the global log-ratio–log-price trend. Note that the direction and interpretation reverse automatically when the fitted trend is progressive.

Note that the direct penalty targets how valuation ratios vary with sale price, but not their overall valuation level. Multiplying every predicted value by the same positive constant shifts all log valuation ratios by the same amount but leaves the covariance $C(f)$ unchanged. Thus, even a model with $C(f) = 0$ can have a median or mean valuation ratio different from one. Hence, valuation level must be evaluated separately.

For second-order models (e.g., LightGBM), the corresponding Hessian of the squared log-price loss plus direct penalty is

$$\nabla_y^2 \left(L(f) + \frac{\rho}{2} C(f)^2 \right) = \frac{2}{n} I + \frac{\rho}{n^2} c c^\top. \quad (19)$$

The rank-one Hessian term $\frac{\rho}{n^2} c c^\top$ introduces cross-observation curvature. Standard LightGBM custom objectives provide one gradient and one Hessian value per observation, so this dense term cannot be represented exactly through the usual interface. An exact second-order implementation therefore requires additional machinery: using only the diagonal is possible, but discards the cross-observation curvature. The next subsection proposes an alternative penalty term to handle this caveat. Further, the model-specific derivations are given in Appendix E.

3.3 Sample-Additive Covariance Upper-Bound Surrogate

For standard second-order boosting software, we seek a penalty that decomposes exactly across observations. Given that (14) already gives $C(f) = \frac{1}{n} \sum_{i=1}^n e_i(f) c_i$, we can make use of Jensen's inequality that yields a sample-additive upper bound to the squared-covariance term directly.

$$C(f)^2 = \left(\frac{1}{n} \sum_{i=1}^n e_i(f) c_i \right)^2 \leq \frac{1}{n} \sum_{i=1}^n e_i(f)^2 c_i^2. \quad (20)$$

This inequality motivates the fully observation-additive surrogate penalty

$$\Psi^{\text{sur}}(f) = \frac{1}{n} \sum_{i=1}^n e_i(f)^2 c_i^2 = \frac{1}{n} \sum_{i=1}^n e_i(f)^2 (y_i - \bar{y})^2. \quad (21)$$

The resulting *surrogate-penalized learner* is therefore

$$f_{\rho}^{\text{sur}} \in \arg \min_{f \in \mathcal{H}} \{J_0(f) + \rho \Psi^{\text{sur}}(f)\}. \quad (22)$$

Thus, $\Psi^{\text{sur}}(f)$ is a genuine upper bound on the squared empirical covariance in (14). Its exact relationship with the direct-covariance penalty can be seen by making the observation

$$\Psi^{\text{sur}}(f) = C(f)^2 + \frac{1}{n} \sum_{i=1}^n (e_i(f) c_i - C(f))^2. \quad (23)$$

Hence, $\Psi^{\text{sur}}(f)$ exceeds $C(f)^2$ by the nonnegative dispersion term in (23): opposite-signed observation-level contributions can cancel in $C(f)$ but not in the surrogate. This upper-bound relation should not be interpreted as control of a broader class of nonlinear residual-price relationships; neither the bound nor its weighted-loss representation directly penalizes nonlinear conditional-mean lack-of-fit.

Combining the baseline loss with the surrogate gives

$$L(f) + \rho \Psi^{\text{sur}}(f) = \frac{1}{n} \sum_{i=1}^n [1 + \rho(y_i - \bar{y})^2] e_i(f)^2. \quad (24)$$

Thus, the surrogate is equivalent to a weighted squared-error objective with fixed observation weights

$$w_i(\rho) = 1 + \rho(y_i - \bar{y})^2. \quad (25)$$

Properties near the center of the log-price distribution receive weights close to the baseline weight, whereas properties in the lower and upper tails receive larger weights.

Since $\exp(\bar{y})$ is the geometric mean sale price, the weight function is symmetric in log-distance from this center: prices that differ from the geometric mean by reciprocal multiplicative factors receive the same additional weight. This symmetry concerns the weighting function only: the empirical lower and upper tails, and the resulting prediction changes, need not be symmetric. Moreover, because the additional weight grows quadratically with log-price distance, observations far from the center can receive disproportionate influence. The log transformation compresses this leverage relative to an analogous raw-price construction, but does not bound it.

This representation clarifies the mechanism without imposing a directional assumption on individual observations. The surrogate raises the cost of residuals farther from the center of the log-price distribution. Unlike the direct penalty, its observation-level correction is not determined by the sign of the aggregate covariance: it depends on the residual $e_i(f)$ and the squared log-price deviation c_i^2 . Its resulting effect on the signed covariance and on assessor-facing vertical-equity diagnostics must therefore be evaluated out of sample.

For squared loss, the observation-wise gradient and Hessian are

$$g_i = \frac{2}{n} e_i(f) [1 + \rho(y_i - \bar{y})^2], \quad (26)$$

$$h_i = \frac{2}{n} [1 + \rho(y_i - \bar{y})^2]. \quad (27)$$

Both quantities depend only on observation i once $\bar{y}$ has been computed. They can therefore be supplied through a standard second-order boosting custom-objective interface. Equation (24) also shows that the same mathematical objective can, in principle, be expressed as weighted squared loss in regression software that supports fixed observation weights. The CCAO paths reported in this paper use the custom-objective implementation below; an alternative native-weighted implementation should not be assumed to reproduce the identical finite boosting path without an explicit implementation-equivalence check.

For the LightGBM implementation, we multiply the complete objective by the common positive factor $n/2$. This leaves the minimizer and the relative weighting indexed by ρ unchanged while putting the $\rho = 0$ derivatives on LightGBM’s native squared-error scale. The supplied surrogate derivatives are therefore

$$g_i = e_i(f) [1 + \rho c_i^2], \quad h_i = 1 + \rho c_i^2.$$

The corresponding direct-objective implementation is given in Appendix E.

The Jensen step is what makes the objective observation-additive, and it is not free. Because the Surrogate multiplies each observation’s squared error by $1 + \rho c_i^2$, the implied weights concentrate on sales far from the centre of the log-price distribution as ρ grows, and the frozen weight audit quantifies how far that goes. At $\rho = 100$ the effective sample size $(\sum_i w_i)^2 / \sum_i w_i^2$ on the development block falls to 86,609 of its 344,607 rows, or 25.1% of n ; across the nine fitting blocks it ranges from 25.0% to 28.6% of n , the median weight on the development block is 19.1 rather than near one, and the most extreme one percent of rows carry 10.7% of the total weight. Weight concentration of this magnitude is an implementation-side diagnostic, and a plausible explanation for the Surrogate’s behaviour at strong penalties. It is an association, not an established mechanism: nothing here demonstrates that the concentration causes the observed path behaviour, and no such claim is made.

Unlike the direct covariance penalty, the surrogate is not invariant to a common shift in all log predictions because it penalizes the magnitude of $e_i(f)$ itself. It can therefore affect overall valuation level in addition to the value-related trend.

3.4 Interpretation, Validation, and Scope

Because LightGBM’s standard custom-objective interface cannot represent the Direct objective’s cross-observation rank-one curvature, we also examined whether the omitted curvature could be treated as negligible over the evaluated configurations. The diagnostic evidence was inconclusive: it did not support a gradient-only interpretation of the fitted Direct objective, but neither did it establish that the omitted curvature materially changes the resulting path. We therefore make neither claim. Appendix E reports the corresponding implementation diagnostics.

What can be stated exactly is the structure. The Hessian of the Direct objective is $\frac{2}{n}I + \frac{\rho}{n^2}cc^\top$, from (19): a diagonal part plus a *dense rank-one* term that couples every pair of observations. LightGBM’s custom-objective interface accepts one gradient and one scalar Hessian per observation, so the cross-observation rank-one term is not representable through it and is omitted. The implementation supplies instead the diagonal of the exactly scaled Hessian: $1 + \frac{\rho}{2n}c_i^2$ for Direct and $1 + \rho c_i^2$ for the Surrogate, with the common factor $n/2$ applied analytically so that the supplied derivatives reduce to $g_i = e_i$ and $h_i = 1$ at $\rho = 0$. The Direct optimizer therefore combines the exact penalty gradient with an approximate, diagonal curvature. The available diagnostics do not establish whether omitting the cross-observation curvature is practically negligible.

The two formulations intervene differently. The Direct penalty responds to the aggregate signed first-order association between log valuation ratios and log sale prices. The Surrogate instead changes the data-fit objective itself by upweighting squared errors farther from the center of the log-price distribution. Although this observation-additive objective upper-bounds squared covariance, it is not merely an implementation approximation to the Direct penalty: its additional dispersion term in (23) gives it a distinct mechanism and can produce different path behavior. Because the two penalties have different scales, the numerical values of ρ are not directly comparable across formulations. Their appropriate scale can also vary across training samples and jurisdictions, so the two regularization paths are traced separately through chronological validation.

Neither formulation directly optimizes PRD, PRB, VEI, COD, or valuation level. Nor does zero training covariance imply the absence of nonlinear, geographic, property-class, temporal, or other local value-related

patterns. The training covariance is therefore treated as a mechanism target rather than as a sufficient measure of vertical equity. Predictive performance, valuation level, PRD, PRB, VEI, ratio curves, the correlation-ratio nonlinearity gap Δ_{NL} in (12), and the broader distance-correlation diagnostic remain separate validation and out-of-sample evaluation criteria; subgroup diagnostics remain necessary complements for later operational validation.

Sale price is required to construct the correction only for verified sales used during model fitting. Once a model has been trained, the resulting valuation function continues to generate predictions from x alone, as in the baseline AVM; the correction does not require a current sale price for an unsold property.

For comparisons across training samples, the raw numerical value of ρ is not intrinsically portable. Under a rescaling $y \mapsto ay$, squared error scales as a^2 while $C(f)^2$ scales as a^4 ; the scale-invariant Direct coordinate is therefore

$$\tilde{\rho} = \rho \widehat{\text{Var}}_{\text{train}}(y). \quad (28)$$

Raw ρ remains the coordinate of the CCAO implementation reported below. Any cross-jurisdiction comparison must recompute the training-block variance without leakage and report $\tilde{\rho}$; even then, normalization is a comparability device, not a universal operating rule.

Section 4 specifies the chronological validation and regularization-path procedure used in the empirical application. Appendix D gives parallel fixed-prediction-space benchmarks for the two penalties. For the Direct penalty, increasing ρ shrinks the targeted covariance smoothly and, when the space contains an intercept, the exact path is a one-parameter centered spread rescaling of the unpenalized fit. For the Surrogate, the exact path is instead a log-price-distance-weighted projection of the outcome onto the same prediction space; locally, it moves in the learnable component of the baseline residuals weighted by squared log-price distance from the sample center. In both cases, the increase in baseline in-sample squared error is second order near $\rho = 0$. These results clarify the rank-one geometry of the Direct correction and the generally multi-directional geometry of the Surrogate, while remaining interpretive benchmarks rather than guarantees for fully retrained boosted trees, whose splits, leaves, and stopping behavior can change with the objective.

4 Application and Evaluation Design

This section specifies the empirical design used to evaluate the proposed correction. We first describe the CCAO application, data, and implementation; then define the chronological validation and regularization-path protocol; and finally summarize the model comparators. All model comparisons reported in the next section use the evaluation criteria defined in Section 2.3.

4.1 The CCAO Residential Pipeline: Application, Data, and Implementation

For the primary application, we implement in Python the components of CCAO’s residential valuation workflow required for model development and evaluation. The implementation follows the data, feature, model-fitting, and ratio-study structure of CCAO’s published residential pipeline, but it is a research translation rather than a direct production run. The reported results should therefore be interpreted as evidence within this translated workflow, not as official CCAO assessment outputs.

Within the LightGBM comparison, the intervention is deliberately restricted to the training objective. We hold fixed the sale filters, log-price target, underlying predictor information, temporal design, transformation back to the price scale, and evaluation procedures. The fixed Section-2 LightGBM hyperparameter configuration is held fixed across the standard and penalized fits, so ρ is the only dimension varied within each penalized family. Holding the surrounding model specification fixed makes the training-objective change as controlled as possible. The native-versus-custom $\rho = 0$ comparison is reported in Appendix E; because the completed parity audit identifies this difference as an execution-path distinction, attribution of positive-penalty changes is made within each custom-objective family.

The primary CCAO application uses verified Cook County residential sales from the final residential-modeling extract used to construct the evaluation samples. The residential scope follows the property types represented in the translated CCAO residential pipeline, including single-family homes, townhomes, small multifamily properties, and condominiums; following the office pipeline, we exclude multicard PINs and

sales marked as outliers. The extract is identified by content rather than by a version string. Every sample reported here is constructed from `data/CCAO/2025/training_data.parquet`, a 215,400,916-byte file with SHA-256

`b1fc00b514041af5aa7135d85ed59a028afe40569bba1014c4dcf3ad2f3a7b51`.

The extract is not redistributed with the replication materials. Its build or retrieval date was not recorded, so none is reported; the content hash identifies the exact file used in the analysis, not when it was produced.

Eligible 2016–2024 sales are ordered chronologically. The oldest 90% (344,607 sales, through 2023-11-09) form the development/CV pool, and the newest 10% (38,290 sales, 2023-11-09 through 2024-12-31) form the primary held-out evaluation. The separate 2025 block contains 26,641 sales and is evaluated after each specification is refit on all 382,897 eligible 2016–2024 sales. Because results from the 2024 block were examined during method development, we treat that block as a held-out evaluation rather than as an untouched or preregistered confirmatory test. For the later lower- ρ resolution extension described in Section 4.2, however, the augmented chronological-CV grid was fixed before the newly added lower-tail points were evaluated on either the held-out block or the 2025 forward sample.

Because the 90/10 split is defined on chronologically ordered observations rather than on distinct calendar dates, sales occurring on the cutoff date can appear on both sides of the split; no sale observation appears in both samples.

Table 3: Final CCAO temporal samples.

Sample	Period	Observations
Development / validation	2016-01-01–2023-11-09	344,607
Primary held-out evaluation	2023-11-09–2024-12-31	38,290
Full 2016–2024 refit sample	2016-01-01–2024-12-31	382,897
Secondary forward evaluation	2025-01-01–2025-12-29	26,641

Observed eligible sales in the final 2016–2024 modeling universe begin on 2016-01-01.

The model uses 95 predictors, including property characteristics, location and proximity variables, Census measures, time variables, parcel-shape measures, neighborhood identifiers, and other assessor variables. Twenty-three are treated as categorical. All predictors are constructed upstream by the CCAO pipeline. Standard and penalized LightGBM fits use the same 95 predictors, with the same 23 variables treated as categorical; the penalty introduces no additional predictors or feature engineering.

Table 4 summarizes the predictor information used by the models. The predictor set is fixed before model fitting and is identical across the standard and penalized fits. It is defined by the restricted extract identified above and is not redistributed as a separate artifact; Appendix H states what is and is not released.

Table 4: CCAO predictor groups used in the empirical models.

Feature group	Count	Examples
Assessor/property variables	30	Class, building area, land area, age, rooms, neighborhood
Location variables	9	Coordinates, census tract, school district, municipality
Proximity variables	26	Transit, parks, roads, hospitals, water, airports
ACS 5-year measures	16	Income, education, poverty, employment, tenure
Time variables	8	Sale year, month, quarter, day, post-COVID indicator
Parcel-shape variables	6	Edge/angle dispersion, bounding ratios, vertex count
Total	95	23 categorical predictors in LightGBM

4.2 Temporal Validation and Regularization-Path Design

Within the 344,607-sale development/CV pool, we use seven expanding-window rolling-origin folds. The cumulative windows advance in 15-month steps anchored at 2016-01-01, and within each cumulative window the newest 10% of observations form validation. For fold v , let $\mathcal{T}_v$ and $\mathcal{V}_v$ denote its training and validation samples. The training window expands across folds, and all training-dependent quantities used by the proposed objectives, including $\bar{y}$ and the centered log-price deviations, are recomputed from $\mathcal{T}_v$ only. The exact fold dates and sample sizes are reported in Appendix F, Table 12; because the newest-10% validation rule is identical in every fold, it is stated once rather than repeated as a table column.

Development coordinates. Three development coordinates appear below, and they are not interchangeable. The primary coordinate, D1, preserves the CCAO-inspired overlapping rolling-origin workflow: it is the equal-weight mean of the seven chronological fold metrics. D3 is the one-sale-one-vote sensitivity, in which every distinct sale carries total weight exactly one through $w_{ik} = 1/m_i$ across the m_i folds in which sale i appears; it is defined on the development pool alone and covers 130,165 distinct sales. D2 is the duplicate-weighted pooled out-of-fold coordinate, retained as a historical and reproducibility sensitivity and never elevated over D1 or D3.

The three differ because of a property of the fold construction that has to be stated before any result is read. The fold-6 and fold-7 origins are only about four months apart, while the rule reserving the most recent tenth of each cumulative window for validation spans close to a year of the development pool, so those two validation blocks overlap. The concatenated pooled sample therefore contains 151,153 appearances of 130,165 distinct sales: 20,988 sales appear twice and none appears more often, which the frozen accounting records as a duplicated share of 13.9%.

D1’s status under that overlap is exactly five things, and the last two are the ones easily lost. D1 remains the primary coordinate; its mathematical definition is unchanged; every fold-level prediction is still genuinely out of training sample, because no overlapping row lies in the training block of the model that predicted it; but D1 is *not* an independent-observation aggregate, since some sales contribute to more than one fold-specific value; and fold-level statistics are therefore not independent. We do not describe D1 as unaffected by the overlap. It follows that the fold-level standard deviations reported alongside fold means are descriptive chronological-window variation and not standard errors: they support no IID reading of the seven folds and no across-fold significance statement. D2 is the only one of the three coordinates that double-counts by construction, and where the nonlinearity gap is computed on that pooled sample a composite `fold|row_id` identifier is supplied to the otherwise unchanged frozen estimator, so the quantity stays a function of observation identity alone.

The experiment traces the full regularization paths rather than choosing a penalty strength at this stage. The substantive comparison contains standard LightGBM and the positive-penalty Direct and Surrogate LightGBM families. Each family was first evaluated on 50 geometrically spaced positive values from 0.1 to 100. A one-time lower-tail extension then added 32 positive values between approximately 0.00110 and 0.08685, aligned to the same logarithmic spacing as the original grid, to increase resolution near zero. The final augmented path therefore contains 82 positive values per family, with the originally fitted 0.1–100 points retained unchanged. Because the direct and surrogate objectives have different numerical scales, their ρ values are not interpreted as comparable magnitudes. The LightGBM hyperparameters were tuned on the same seven chronological folds before any penalty path was estimated. The tuning criterion was the equal-weight mean of fold validation price RMSE, $\text{RMSE}_P(S) = \sqrt{n_S^{-1} \sum_{i \in S} (\hat{P}_i - P_i)^2}$. The adopted configuration uses 994 trees and was then frozen for ordinary LightGBM, Direct, and Surrogate fits. The held-out and 2025 samples were not used for this hyperparameter choice. With that vector held fixed, ρ is the only dimension varied within each penalized family. The complete parameter vector is archived in the frozen preflight provenance together with its configuration hash, which is re-verified there against the executed configuration file and against both experiment specifications; that hash is quoted in Appendix E, so the main text does not list every parameter.

For each configuration and fold, we retain the complete set of predictive, valuation-level, ratio-dispersion, vertical-equity, and mechanism/residual-structure diagnostics. We report equal-weight fold means and standard deviations to describe the chronological behavior of the regularization paths. No deployment operating point is selected in the present analysis. The chronological-CV path is used only to characterize the path.

Table 5: CCAO application and regularization-path design.

Item	Design
Population	Verified Cook County residential sales
Development / CV	Oldest 90% of eligible 2016–2024 sales: 344,607, through 2023-11-09
Primary held-out evaluation	Newest 10%: 38,290 sales, 2023-11-09–2024-12-31
Full 2016–2024 refit sample	382,897 eligible 2016–2024 sales
2025 forward evaluation	26,641 sales, 2025-01-01–2025-12-29
Predictors	95 total; 23 categorical
Validation	Seven expanding-window rolling-origin folds; newest 10% validates
Base nonlinear learner	LightGBM on log sale price; 994 trees; hyperparameters frozen before the penalty sweep
Corrections evaluated	Direct squared covariance and sample-additive covariance upper bound
Penalty path	82 positive values per family: original 50-point [0.1, 100] path plus a 32-point lower-tail extension [0.00110, 0.08685]
Current analysis	Full CV, held-out, and forward paths; descriptive path landmarks only; no screening result and no deployment operating-point selection

For compact presentation, we use five nominal decade-spaced positive display anchors, $\rho \in \{0.01, 0.1, 1, 10, 100\}$, and represent each nominal value by the nearest already-tested point on the frozen 82-point positive grid. The corresponding tested values are 0.0104811, 0.1, 0.954095, 10.4811, and 100. This display convention is adopted only to make the tables and ratio-shape figures easy to read; it is not part of the frozen transition rule and does not select a penalty strength. Ordinary LightGBM remains the unpenalized baseline.

All 82 positive values per family remain in the final path figures; no tested value is dropped from the reported paths.

Ratio-shape figures additionally show the zero-penalty path origin for visual continuity. Because the Direct and Surrogate objectives have different numerical scales, equal nominal display anchors are common presentation coordinates only and must not be interpreted as equal-strength interventions across families.

The same final augmented regularization paths are evaluated on the two out-of-time samples. For the held-out evaluation, every valid configuration is refit on the full 344,607-sale development pool and evaluated on the 38,290-sale held-out sample. For the 2025 forward evaluation, the same configuration is refit on all 382,897 eligible 2016–2024 sales and evaluated on the 26,641-sale 2025 sample. These held-out and forward paths are descriptive. For the v2 lower-tail extension, the augmented grid and its CV-derived transition result were frozen before the newly added lower- ρ points were evaluated out of time; those new held-out and forward results were not used to revise the grid, the transition rule, the objective definitions, or the frozen LightGBM configuration. The chronological CV, held-out, and 2025 paths are therefore complementary temporal views of the final augmented experiment, not sequential stages that select a penalty strength.

All scalar evaluation statistics are computed on the full relevant validation or evaluation sample after predictions are transformed back to the price scale. For the ratio-shape figures in the Results, each evaluation sample is partitioned into 30 equal-count sale-price bins and we plot the median valuation ratio in each bin against sale price on a logarithmic horizontal axis. The binning is used only for visualization; all scalar metrics are computed on the full evaluation sample.

4.3 Models and Comparisons

The empirical comparison in this paper covers the baseline model class, the custom-objective implementation control, and the Direct and Surrogate training-time regularization paths, together with a global post-hoc centered rescaling of the same unregularized predictors. That comparator is reported in Sections 5.4 and 5.5, both descriptively and at matched development first-order correction. *Structured* post-hoc calibration by

geography, property class or time is a different construction and remains outside the empirical scope.

Reference cells, and what each one licenses. Three reference cells appear below, and the distinction between two of them is load-bearing for every comparison in the Results. *Ordinary LightGBM (standard raw-label native)* is the assessor-facing workflow benchmark: the standard pipeline as an office would run it. *Centered-label native L2 (initialization-aligned)* is an implementation-decomposition control only, and no full regularization path is traced for it. *Custom-objective $\rho = 0$ origin* is the primary within-path penalty-isolating reference: the same custom-objective code path as the penalized fits, evaluated at zero penalty.

The attribution rule that goes with them is not a nicety. Comparisons against the workflow benchmark are descriptively valid, and they are the practitioner’s question – can a different objective beat the standard pipeline? – but a change relative to it must not be attributed to ρ alone, because that cell differs from the penalized fits in execution path as well as in penalty. The custom-objective $\rho = 0$ origin to positive- ρ contrast, taken within the same family, is therefore the primary within-path penalty-isolating comparison: it isolates the change associated with activating the penalty within the same frozen implementation path. Where both comparisons are reported they are labelled as the different questions they are, and no workflow-benchmark difference is presented as a pure penalty effect. Appendix E reports the $\rho = 0$ implementation audit that makes the distinction necessary rather than merely cautious.

The workflow benchmark. Standard LightGBM, trained with the ordinary squared log-price objective, is the workflow benchmark for every comparison reported here; the penalized and post-hoc configurations use the same outcome, development observations, and underlying predictor information under the common temporal design. The earlier township-level linear-regression workflow is historical context only: it is not re-estimated, and no quantitative linear-versus-boosted comparison is reported.

Direct and surrogate penalties. The direct LightGBM comparator uses the diagonal-Hessian approximation described in Appendix E: its covariance gradient is exact, while the cross-observation second-order curvature is omitted so that the objective can be used through the standard LightGBM interface. The surrogate uses the exact observation-additive objective from Section 3.3.

Post-hoc centered rescaling. A global one-parameter post-hoc centered rescaling of an already fitted predictor is *inside* the empirical scope of this paper. The fixed-space Direct result in Appendix D makes that comparison scientifically natural, and Sections 5.4 and 5.5 carry it out: first descriptively, then at matched development first-order correction, with the rescaling of the custom-objective $\rho = 0$ origin as the primary post-hoc family and the rescaling of the workflow benchmark as a secondary practical variant. The paper’s contribution remains the design, implementation, mechanism analysis and evaluation of training-time covariance-guided corrections; the comparator is what keeps that contribution from resting on an untested assumption that retraining is necessary.

In many settings, richer spatial, temporal, neighborhood, and property information can improve both predictive accuracy and assessor-facing equity measures [49], but that is a modeling extension rather than a comparator in the headline experiment. We therefore discuss richer information as a separate direction in Section 6 rather than treating it as a completed empirical control here.

4.4 Scope of External Evidence

The empirical evidence in this paper is CCAO-specific. We therefore make no claim that the same penalty scale, attainable correction, or accuracy–equity path transfers to other jurisdictions. Establishing such transfer would require evaluation under jurisdiction-specific data, legal rules, feature engineering, and assessor review procedures.

5 Results

The figures below show the complete Direct and Surrogate penalty paths. Table 6 reports prediction, PRD, and MKI where the positive- ρ values can be reconstructed directly from retained fitted artifacts; Table 7

reports PRB and VEI at all five display anchors from the retained inferential tables. The anchors illustrate the paths; they are not selected models.

5.1 Baseline and Reference Fits

Ordinary LightGBM predicts sale prices well, but its valuation ratios decline with sale price (Table 2 and Figure 1). On held-out sales, $R_P^2 = 0.894$, PRB is -0.091 , and VEI is -26.5% ; the pattern also appears in 2025. These values describe the unpenalized workflow benchmark, not a comparison between model classes.

The Direct and Surrogate custom-objective fits coincide at $\rho = 0$, but this shared fit differs numerically from ordinary LightGBM (Appendix E.3). We therefore compare each positive- ρ fit with the shared custom-objective origin to describe the change along that penalty path. Ordinary LightGBM remains the workflow benchmark; a difference from it also includes the native-versus-custom implementation difference.

5.2 Regularization Paths and Ratio Shape

Tables 6 and 7 give snapshots of the full 82-positive-value path for each family. Because the penalties have different scales, the same numerical ρ is not the same correction strength in the two families. The tables retain different numbers of anchors because their reported metrics have different reconstruction coverage.

Table 6: Prediction, PRD, and MKI at the two positive penalty anchors that can be reconstructed directly from retained fitted artifacts. Both reference fits and both later evaluation samples are shown.

Method	ρ	R_P^2	MAE	MAPE	RMSE _{log P}	PRD	MKI
<i>Panel A: Held-out evaluation</i>							
Reference cells							
Ordinary LightGBM (standard raw-label native)	—	0.894	\$75,655	21.2%	0.289	1.069	0.923
Custom-objective $\rho = 0$ origin	0	0.893	\$75,976	21.2%	0.289	1.069	0.923
Direct penalty							
Direct	≈ 0.954095	0.899[†]	\$74,485[†]	21.1%[†]	0.290	1.060[†]	0.940[†]
Direct	100	0.869	\$84,918	23.9%	0.325	1.029[†]	0.998[†]
Surrogate penalty							
Surrogate	≈ 0.954095	0.897[†]	\$75,468[†]	21.0%[†]	0.298	1.049[†]	0.953[†]
Surrogate	100	0.889	\$82,094	22.8%	0.357	1.010[†]	1.013[†]
<i>Panel B: 2025 forward evaluation</i>							
Reference cells							
Ordinary LightGBM (standard raw-label native)	—	0.904	\$78,484	20.8%	0.278	1.079	0.907
Custom-objective $\rho = 0$ origin	0	0.904	\$79,139	20.8%	0.279	1.078	0.909
Direct penalty							
Direct	≈ 0.954095	0.910[†]	\$77,139[†]	20.7%[†]	0.279 [†]	1.069[†]	0.926[†]
Direct	100	0.873	\$89,309	23.1%	0.307	1.036[†]	0.983[†]
Surrogate penalty							
Surrogate	≈ 0.954095	0.908[†]	\$78,196[†]	20.4%[†]	0.284	1.060[†]	0.932[†]
Surrogate	100	0.900	\$85,257	22.0%	0.336	1.020[†]	0.987[†]

Notes. Ordinary LightGBM is the workflow benchmark; the custom-objective $\rho = 0$ fit is the within-path reference. Bold indicates improvement over the workflow benchmark and a dagger ([†]) improvement over the custom-objective origin, each in the preferred direction of the metric using unrounded values. Marks do not imply statistical significance, compliance, or selection. Only $\rho \approx 0.954095$ and 100 can be reconstructed directly for these measures; Table 7 gives PRB and VEI at five anchors. Penalty magnitudes are not comparable across families. PRD has an adopted $[0.98, 1.03]$ guidance range; the MKI $[0.95, 1.05]$ band is proposed in the May-2026 Exposure Draft. MAE is in dollars and MAPE in percent.

At the displayed anchor $\rho \approx 0.954095$, Direct improves held-out R_P^2 from 0.893 at its custom-objective origin to 0.899 and reduces MAE from \$75,976 to \$74,485; PRD moves from 1.069 to 1.060. The same configuration improves price-scale fit in 2025. The Surrogate also improves held-out price-scale fit at this anchor, while its PRB moves from -0.088 to -0.045 and VEI from -26.2% to -12.6% . At $\rho = 100$, held-out

Direct R_P^2 falls to 0.869 and MAE rises to \$84,918 even as its PRD approaches one. Stronger correction can therefore improve one diagnostic while worsening another.

Table 7: PRB (adopted 2013 Standard) and VEI (proposed May-2026 Exposure Draft) at five display anchors, on held-out and 2025 evaluations. Both reference fits are shown.

		PRB (adopted 2013 Standard)		VEI (May-2026 Exposure Draft, proposed)	
Method	ρ	Held-out	2025 forward	Held-out	2025 forward
Reference cells					
Ordinary LightGBM (standard raw-label native)	–	-0.091	-0.106	-26.5%	-28.6%
Custom-objective $\rho = 0$ origin	0	-0.088	-0.103	-26.2%	-28.4%
Direct penalty					
Direct	≈ 0.0104811	-0.089	-0.105	-26.9%	-29.8%
Direct	0.1	-0.088[†]	-0.102[†]	-27.0%	-27.7%[†]
Direct	≈ 0.954095	-0.075[†]	-0.090[†]	-21.9%[†]	-23.9%[†]
Direct	≈ 10.4811	-0.040[†]	-0.049[†]	-10.5%[†]	-11.9%[†]
Direct	100	-0.015[†]	-0.029[†]	0.7%[†]	-4.5%[†]
Surrogate penalty					
Surrogate	≈ 0.0104811	-0.088[†]	-0.101[†]	-25.8%[†]	-27.4%[†]
Surrogate	0.1	-0.079[†]	-0.095[†]	-22.7%[†]	-24.7%[†]
Surrogate	≈ 0.954095	-0.045[†]	-0.062[†]	-12.6%[†]	-15.3%[†]
Surrogate	≈ 10.4811	0.013[†]	0.000[†]	0.2%[†]	-2.0%[†]
Surrogate	100	0.042[†]	0.030[†]	4.0%[†]	1.7%[†]

Notes. All PRB and VEI entries are reconstructible from retained inferential tables. Bold indicates movement toward zero relative to ordinary LightGBM; a dagger ([†]) indicates movement relative to the custom-objective $\rho = 0$ origin, using unrounded values. Neither mark implies significance, compliance, or selection. PRB has an adopted $[-0.05, 0.05]$ range; the VEI $\pm 10\%$ band is proposed. The Direct held-out VEI of -10.5% near $\rho = 10$ lies outside that proposed band without establishing a formal finding. The Surrogate 2025 PRB printed as 0.000 at that anchor is 0.000135 before rounding (95% CI $[-0.004, 0.005]$). Penalty magnitudes are not comparable across families.

Valuation-ratio profiles. Figure 2 shows where the ratios change across sale prices. Both families reduce the broad downward trend. At stronger penalties the Surrogate develops a trough in the lower and middle price range; Direct changes more smoothly in these displayed profiles. A near-zero slope or grouped statistic does not imply a flat ratio profile.

First-order slope. Along the held-out path, Direct’s $\beta_{\log}$ moves from -0.147 at $\rho = 0$ to -0.079 at $\rho = 100$, while the Surrogate moves from the same origin to -0.001 . The corresponding 2025 endpoints are -0.161 to -0.093 for Direct and -0.161 to -0.015 for the Surrogate (Figure 3). The path-anchor artifact does not carry reconstructible positive- ρ Δ_{NL} or dCor columns, so the numerical comparisons of those measures below use separately retained matched-target records.

5.3 Accuracy and Equity Together

Figure 4 plots price-scale fit against four assessor-facing vertical-equity measures over the tested paths. At moderate penalties, portions of both paths improve vertical-equity diagnostics without lowering R_P^2 relative to the shared custom-objective origin. At stronger penalties, R_P^2 falls as some diagnostics approach or pass their neutral values. The full paths show these changes more clearly than any single anchor.

5.4 Post-hoc Rescaling of the Unpenalized Fit

The fixed-prediction-space Direct result motivates a simple comparator: rescale the shared $\rho = 0$ predictor around its fitting-block mean log sale price, $f_b(x) = \bar{y}_T + b(f_0(x) - \bar{y}_T)$, without refitting. The scale factor is determined from development predictions. Appendix F.5 reports the full construction, its sensitivity checks, and a secondary rescaling of ordinary LightGBM.

Rescaling can move the first-order slope substantially, but pushing the development slope to zero requires $b = 1.160458$. At that value, held-out R_P^2 falls from 0.8928 to 0.8063, MAE rises from \$75,976 to \$87,182, and Δ_{NL} rises from 0.1164 to 0.1343 (Appendix Table 14). This comparator shows that the first-order trend can be moved without retraining; the next comparison tests the methods at similar amounts of movement.

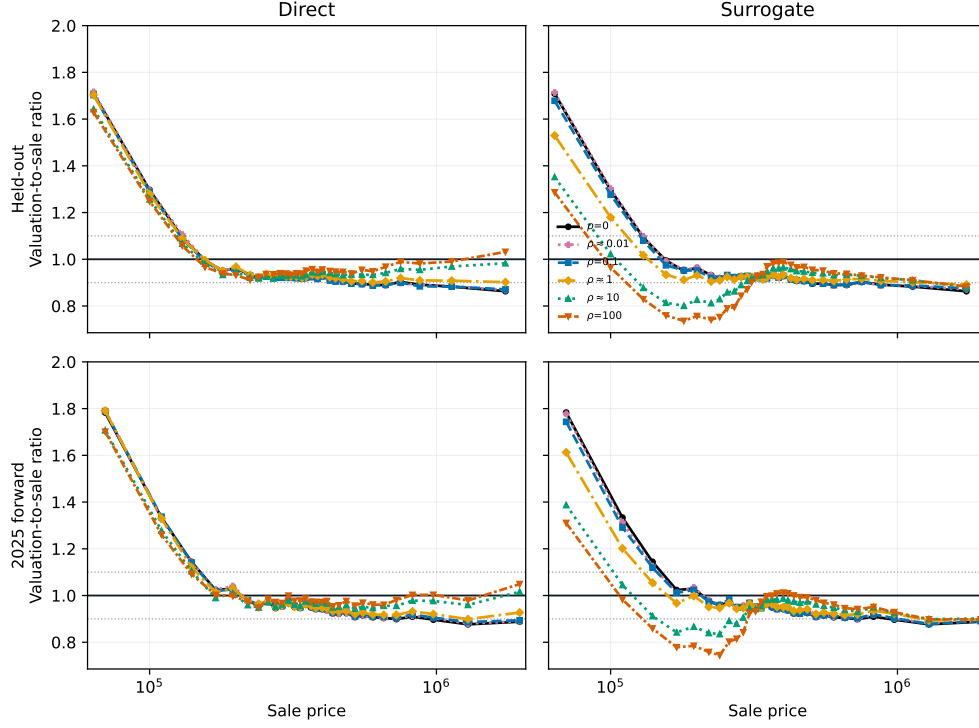


Figure 2: Median valuation-to-sale ratios across 30 equal-count sale-price bins for the shared $\rho = 0$ fit and five display anchors, on held-out and 2025 sales. The line at 1 denotes equal valuation and sale price; the 0.9 and 1.1 guides are aggregate level references, not binwise criteria.

5.5 Comparison at Matched First-Order Correction

We match Direct, Surrogate, and post-hoc rescaling at six targets within the first-order correction range they all attain on the development folds (Appendix Table 15). The match is on the development slope; it does not equalize slopes on held-out or 2025 sales. The uncorrected target is an identity check because all three paths start from the same fit.

At the strongest shared development target, Direct has the lowest held-out R_P^2 and the highest MAE, log-price RMSE, and COD of the three (Table 8). Across all five nonzero shared targets, the Surrogate has a lower Δ_{NL} than both alternatives (Appendix Table 15). Post-hoc rescaling reproduces much of Direct’s first-order tradeoff over the range Direct reaches. These are differences between approximately matched configurations, not estimates at identical held-out slopes.

Table 8: Held-out results at the strongest development first-order correction attained by all three families ($j = 5$ in Appendix Table 15). Matching is on development predictions; held-out first-order slopes need not coincide.

Family	Configuration	R_P^2	MAE	$RMSE_{\log P}$	COD	Δ_{NL}	dCor
Direct	$\rho = 86.8511$	0.8760	\$82,764	0.3170	23.97	0.1198	0.2730
Surrogate	$\rho = 2.5595$	0.8963	\$76,128	0.3094	21.40	0.0914	0.2714
C-posthoc	$b = 1.069186$	0.8966	\$74,236	0.2920	21.61	0.1282	0.2944

Notes. C-posthoc rescales the shared custom-objective $\rho = 0$ fit without retraining. The Direct and Surrogate penalties have different numerical scales. COD is in percent and remains above the adopted [5, 15] reference range; Δ_{NL} and dCor have no assessor guidance ranges. Price-scale accuracy here uses direct exponentiation; the retransformation check is in Appendix F.7.

The price-scale ranking is sensitive to how log predictions are returned to dollars: the Surrogate-versus-

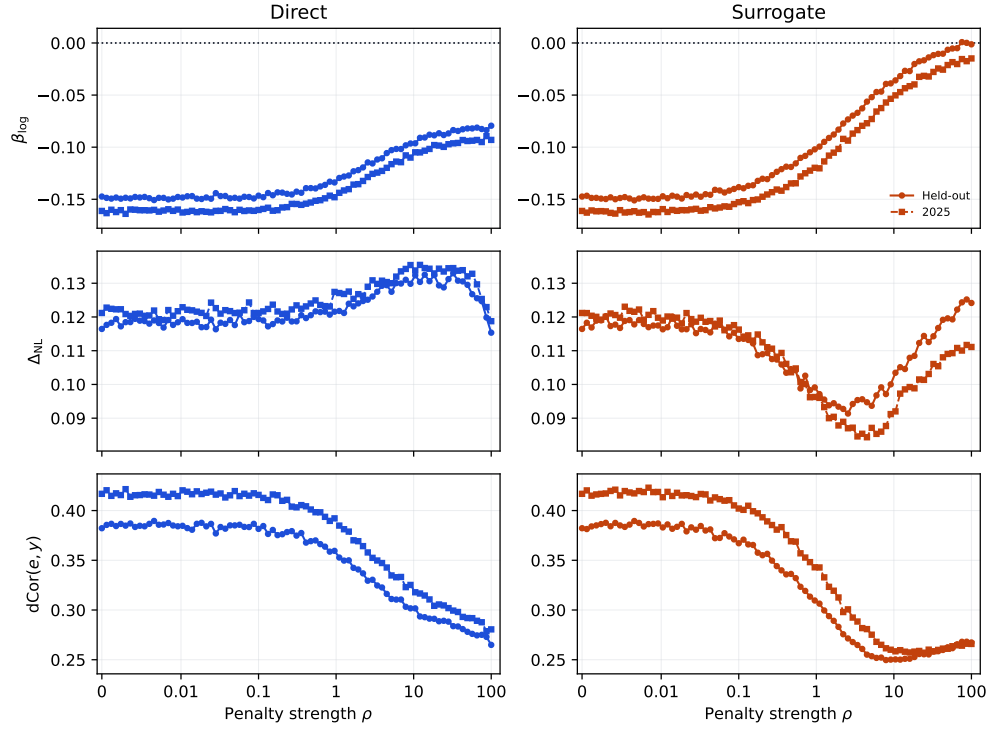


Figure 3: First-order slope $\beta_{\log}$ and descriptive residual-structure paths on held-out (solid) and 2025 (dashed) sales. Only the $\beta_{\log} = 0$ line marks a first-order neutrality target. The retained path-anchor artifact does not itself reproduce positive- ρ Δ_{NL} or $dCor$ values; their supported positive- ρ comparisons in the text use the separate matched-target records.

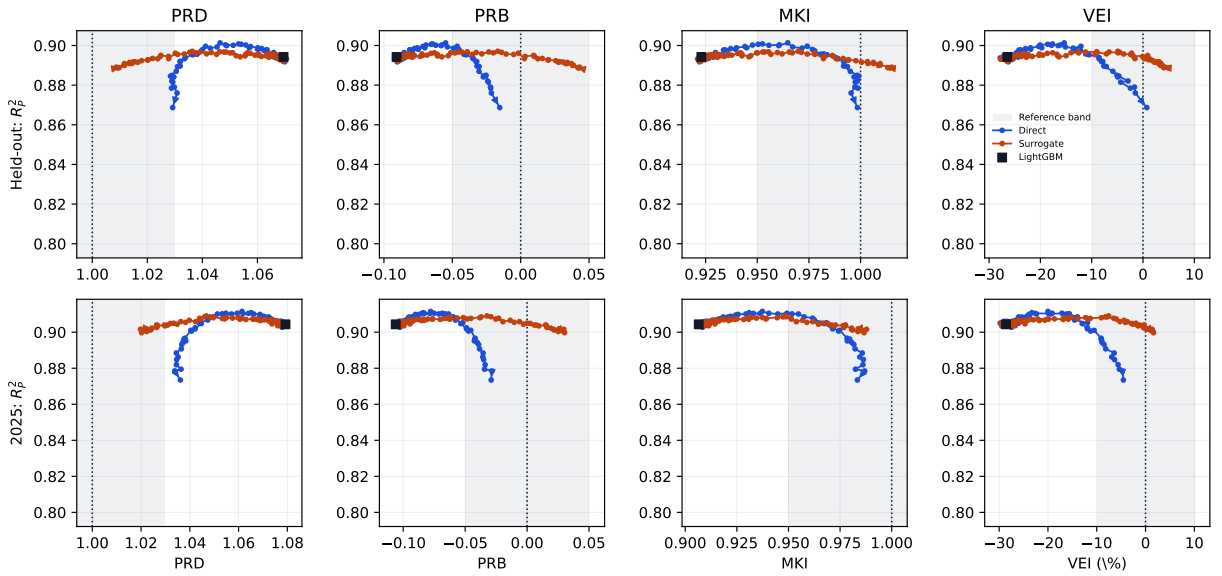


Figure 4: Predictive fit against PRD, PRB, MKI, and VEI across the held-out and 2025 paths. The square is ordinary LightGBM; arrows show increasing ρ . Shading shows assessor-facing reference bands, including proposed bands for MKI and VEI, without a compliance or selection claim.

post-hoc ordering in R_P^2 and MAE at this target reverses under the recorded smearing check (Appendix F.7). Accuracy can also rank the families differently across scales. At the extended development target $\beta_{\log} = -0.03$, Surrogate and post-hoc have held-out R_P^2 of 0.8917 and 0.8569, respectively, but log-price RMSE of 0.3443 and 0.2999. Direct did not reach this target on the evaluated grid; the Surrogate did not reach zero on the development coordinate. Appendix Table 16 reports these attainment limits. Reweighting the overlapping development folds by distinct sale changes some near-tie rankings but leaves the matched Δ_{NL} ordering and Direct’s degradation at the strongest shared target intact (Appendix F.6).

5.6 Chronological Stability

The seven expanding-window validation folds support a directional change in the first-order slope, but their preferred metric locations do not provide a stable penalty strength for later periods. The final two validation blocks overlap, so fold-to-fold variation describes chronological windows rather than independent standard errors. On the Surrogate path, the held-out slope rises slightly above zero, whereas the chronological-CV mean and the 2025 forward slope remain negative. The same penalty therefore does not produce the same first-order result in each period. No portable operating interval or deployment penalty is identified.

Strict-date, parcel-overlap, and unseen-parcel checks do not materially contradict these within-path findings after local near-zero ties are resolved. Their definitions, limitations, and results are retained in Appendix F.

6 Discussion and Conclusions

The ordinary LightGBM workflow benchmark predicts sale prices well, yet its valuation ratios decline across the price range on verified sales. This result motivates a targeted correction; it does not show that machine learning itself causes regressivity. The Direct and Surrogate objectives target the global first-order log-ratio/log-sale-price trend. Along parts of both paths, this trend falls without a loss of price-scale R_P^2 relative to their shared custom-objective $\rho = 0$ fit. Stronger penalties can instead worsen accuracy, valuation level, dispersion, or nonlinear ratio patterns. Ordinary LightGBM remains the workflow benchmark, but the custom-objective $\rho = 0$ fit is the reference that isolates each penalty’s effect; the two unpenalized implementations differ numerically (Appendix E.3).

The alternatives matter as much as the training paths. Direct penalizes the sample-wide covariance; its LightGBM implementation supplies the exact gradient and diagonal curvature. Surrogate is observation-additive and changes the weight placed on prediction errors across the sale-price range. A simple centered post-hoc rescaling of the unpenalized predictions reproduces much of Direct’s first-order movement without retraining (Sections 5.4 and 5.5). At the reported nonzero matched development slopes, the Surrogate leaves a smaller nonlinearity gap than the two comparators, while Direct has the weakest accuracy and dispersion at the strongest target all three attain. These are comparisons at specified targets, not a ranking across the complete paths. The fixed-prediction-space analysis explains why Direct and Surrogate need not make the same change, but it does not predict the exact behavior of retrained trees (Appendix D).

One price-scale comparison also depends on retransformation. Predictions in the main analysis are returned to dollars by direct exponentiation; in the recorded smearing sensitivity, the Surrogate-versus-post-hoc ordering in price-scale R_P^2 and MAE reverses at a matched target (Appendix F.7). Ratio-uniformity and residual-structure statistics are unchanged by a common multiplicative rescaling of predictions, whereas overall valuation level changes. The price-scale result at that target therefore does not establish general predictive superiority for either method.

The paths also change across market periods. The seven chronological folds support the direction of the targeted correction, but the final two validation blocks overlap and are not independent replications. Development, held-out, and 2025 forward results do not identify a stable penalty strength or a portable operating interval (Section 5.6). This temporal variation is a substantive limit on the proposed correction, not a reason to choose a favorable point from the reported paths.

6.1 Limitations

Sale price is both the prediction target and part of the valuation ratio. It is an imperfect proxy for latent market value; unpredictable transaction variation can produce a negative residual–sale-price association even under a well-fitted model (Appendix B). Moving that association toward zero is therefore not, by itself, proof of improved equity relative to unobserved market values. Neither objective directly optimizes PRB, VEI, valuation level, nonlinear ratio structure, or local equity.

The evidence comes from verified sales in a Python research translation of CCAO’s published residential workflow. The LightGBM configuration is held fixed at 994 trees to study the objectives, so the paths are not fully retuned frontiers. Sales need not represent unsold properties, and countywide patterns can hide differences across neighborhoods and property classes. Recent Cook County evidence likewise finds that assessment regressivity differs across property classes [9]. The external jurisdiction analyses are exploratory; the CCAO paths do not establish results for other counties. These are not official assessments or estimates of final tax bills.

The reported path differences are primarily descriptive. Overlapping chronological folds do not yield independent standard errors, and the main analysis does not provide inferential uncertainty for every comparison. Retransformation affects some price-scale rankings, as noted above. No prospective rule for selecting a penalty strength has been validated, so this study does not recommend an operational setting.

6.2 Future Work

The most useful next comparison would test the current baseline, richer property, spatial, and temporal information, the correction, and their combination under the same chronological design. Structured corrections by neighborhood, property class, or period should compare training-time and post-hoc approaches on the same future sales, with uncertainty estimates and subgroup ratio profiles. A deployment study would also need assessor review and a selection rule fixed before evaluating new periods. Appendix G retains a more detailed checklist for that later work.

6.3 Conclusion

In this CCAO application, the correction provides one way to reduce a global price-related pattern in valuations on observed sales. The post-hoc comparison shows that much of Direct’s first-order correction is possible without retraining, while the remaining differences and temporal instability make a single preferred method or deployment penalty unsupported by these results.

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A Metric Details and Guidance Status

A.1 Accuracy and supporting ratio-study formulas

For an evaluation sample S , let $n_S = |S|$ and $\bar{P}_S = n_S^{-1} \sum_{i \in S} P_i$. The predictive measures used in the main text are

$$\text{MAE}_P(S) = \frac{1}{n_S} \sum_{i \in S} |\hat{P}_i - P_i|, \quad (29)$$

$$\text{MAPE}_P(S) = 100 \times \frac{1}{n_S} \sum_{i \in S} \left| \frac{\hat{P}_i - P_i}{P_i} \right| = 100 \times \frac{1}{n_S} \sum_{i \in S} |r_i - 1|, \quad (30)$$

$$R_P^2(S) = 1 - \frac{\sum_{i \in S} (\hat{P}_i - P_i)^2}{\sum_{i \in S} (P_i - \bar{P}_S)^2}, \quad (31)$$

$$\text{RMSE}_{\log P}(S) = \left(\frac{1}{n_S} \sum_{i \in S} e_i(f)^2 \right)^{1/2}. \quad (32)$$

The coefficient of variation (COV) is

$$\text{COV}(S) = 100 \times \frac{s_{r,S}}{\bar{r}_S}, \quad s_{r,S} = \left(\frac{1}{n_S - 1} \sum_{i \in S} (r_i - \bar{r}_S)^2 \right)^{1/2}. \quad (33)$$

For the Modified Kakwani Index (MKI), let $\pi(1), \dots, \pi(n_S)$ index the observations in S in nondecreasing order of sale price, and let

$$q_j = \frac{j - \frac{1}{2}}{n_S}, \quad \bar{\bar{P}}_S = \frac{1}{n_S} \sum_{i \in S} \hat{P}_i.$$

The concentration index of predicted values ordered by sale price and the Gini coefficient of sale prices are

$$C_{\hat{P}|P}(S) = \frac{2}{n_S \bar{\bar{P}}_S} \sum_{j=1}^{n_S} \hat{P}_{\pi(j)} q_j - 1, \quad G_P(S) = \frac{2}{n_S \bar{P}_S} \sum_{j=1}^{n_S} P_{\pi(j)} q_j - 1. \quad (34)$$

Provided there is sufficient sale-price variation, the index is

$$\text{MKI}(S) = \frac{C_{\hat{P}|P}(S)}{G_P(S)}. \quad (35)$$

A.2 Assessor-facing guidance status

The 2013 IAAO Standard on Ratio Studies is the established standard cited for PRD and COD guidance in this paper [27]. The May 2026 revision of the ratio-studies document is an exposure draft. Measures introduced or revised there, including MKI and VEI, are reported as descriptive assessor-facing diagnostics rather than as final adopted compliance standards [32]. Reference bands are used as descriptive guidance in the present regularization-path analysis; the paper does not infer legal compliance from a point estimate.

Value-proxy convention and executable check. The 2013 IAAO Standard defines the equal-weight market-value proxy as $0.50(\text{AV}/\text{Median}) + 0.50 \text{ SP}$. The May 2026 exposure draft describes an equal-weight proxy in its VEI appendix but displays the normalized valuation term without an explicit 0.50 multiplier. All PRB and VEI results here use the single equal-weight convention in Eq. (8), consistent with the 2013 Standard and the draft’s stated interpretation. The executed metric code in `utils/motivation_utils.py` computes $0.5 \text{ SP} + 0.5 (\text{AV}/\text{Med})$ for PRB and $0.5 y_{\text{true}} + 0.5 (y_{\text{pred}}/m_S)$ for VEI. Both match Eq. (8). The SHA256 of that file at population time is 4219a9d68e5266e300b319660f93d37ca184189ee340edb0785f40c9f3c39beb.

The weighted mean ratio satisfies

$$\bar{r}_W = \frac{\sum_i r_i P_i}{\sum_i P_i} = \frac{\sum_i \hat{P}_i}{\sum_i P_i},$$

and PRD is $\bar{r}/\bar{r}_W$. PRB follows the IAAO median-normalized regression definition. The implementation executed for this application is the metric code tracked in the repository, and the transformations, trimming rules and proxy definition it applies are fixed in the frozen conventions catalogued in Appendix H.

A.3 Mechanism and residual-structure diagnostics

The three diagnostics introduced after the assessor-facing measures play distinct roles. The sample slope $\beta_{\log}(S)$ in Eq. (11) is the direct out-of-sample analogue of the covariance mechanism: on a fixed evaluation sample it differs from empirical covariance only by the fixed factor $\widehat{\text{Var}}_S(y)^{-1}$.

For the nonlinear diagnostic, let

$$Z = \frac{y - \mathbb{E}[y]}{\sqrt{\text{Var}(y)}}, \quad m(z) = \mathbb{E}[e \mid Z = z],$$

and let $\Pi_{\text{aff}} m$ denote the $L^2(P_Z)$ projection of m onto the affine span of $\{1, Z\}$. Pearson’s correlation ratio, also called the nonparametric coefficient of determination, is

$$\eta^2(e \mid Z) = \frac{\text{Var}(m(Z))}{\text{Var}(e)} = \frac{\text{Var}(\mathbb{E}[e \mid Z])}{\text{Var}(e)}$$

[19]. Because Z is standardized, the affine component explains the fraction $\text{Corr}(e, Z)^2$ of the variance of e . Orthogonal projection therefore gives

$$\Delta_{\text{NL}} = \eta^2(e \mid Z) - \text{Corr}(e, Z)^2 = \frac{\mathbb{E}[\{m(Z) - (\Pi_{\text{aff}} m)(Z)\}^2]}{\text{Var}(e)} \geq 0. \quad (36)$$

Thus Δ_{NL} isolates the conditional-mean explanatory power that is not captured by the best affine residual-price relationship. We use *correlation-ratio nonlinearity gap* as a descriptive name and Δ_{NL} as paper-specific shorthand; neither the name nor the symbol is claimed as canonical notation from Doksum and Samarov [19]. That paper provides the correlation-ratio/nonparametric- R^2 foundation and explicitly notes its use in constructing measures of regression nonlinearity. The integrated squared-difference goodness-of-fit framework of Härdle and Mammen [22] is closely related to the projection representation above, but it is not the source of the exact normalized statistic in Eq. (36).

For the empirical CCAO analysis, the finite-sample implementation is fixed before the penalized paths are compared. Each held-out or forward evaluation sample S is assigned deterministically to five folds using sale identifiers. For each fold, we fit on the complementary observations (i) an affine regression of e on Z and

(ii) a cubic B -spline regression of e on Z with eight quantile knots, and we score both fits on the omitted fold. Let $\text{MSE}_{\text{aff}}(S)$ and $\text{MSE}_{\text{spl}}(S)$ denote the resulting pooled out-of-fold mean squared errors. The reported sample statistic is

$$\hat{\Delta}_{\text{NL}}(S) = \max \left\{ 0, \frac{\text{MSE}_{\text{aff}}(S) - \text{MSE}_{\text{spl}}(S)}{\widehat{\text{Var}}_S(e)} \right\}. \quad (37)$$

The nonnegative truncation respects the population inequality in Eq. (36); finite-sample cross-fitting can otherwise produce a small negative difference when the flexible fit does not improve out-of-fold squared error. For readability, the empirical tables use Δ_{NL} as shorthand for $\hat{\Delta}_{\text{NL}}(S)$. The estimator is reported on the primary held-out and 2025 forward paths. The same frozen cross-fitted estimator was later applied, fold by fold, to the already-stored chronological validation predictions; those CV values are a retrospective diagnostic and were not used to define the ρ grid or any model-selection rule.

Finally, because $e = \log r$ and $y = \log P$,

$$\text{dCor}(e, y) = \text{dCor}(\log r, \log P). \quad (38)$$

Population distance correlation is zero if and only if independence under the usual moment conditions [52]. The paper uses its sample analogue only as an omnibus diagnostic of residual–price dependence; unlike $\beta_{\log}$ it is unsigned, and unlike Δ_{NL} it is not restricted to conditional-mean structure. It has no assessor-standard reference range.

Every reported value comes from one implementation: the `distance_correlation` routine of the `dcor` Python package, called on the full evaluation sample with the package’s default unit exponent and without bias correction, so the executed estimator is the usual biased distance correlation computed from double-centered distance matrices, rather than the U -centered bias-corrected squared-distance-correlation estimator the package also exports but that this paper never calls. The executed call passes the signed log residual $e = \log \hat{P} - \log P$ as its first argument and $y = \log P$ as its second.

B Squared-Error Prediction and Residual–Value Dependence

Let $Y = \log P$, let X be the observed feature vector, and let

$$f^*(X) = \mathbb{E}[Y \mid X]$$

be the squared-error Bayes predictor. Define $e^* = f^*(X) - Y$. If $\mathbb{E}[Y^2] < \infty$, then

Proposition 1.

$$\text{Cov}(e^*, Y) = -\mathbb{E}[\text{Var}(Y \mid X)] \leq 0.$$

Proof. Using the law of total variance,

$$\text{Cov}(e^*, Y) = \text{Cov}(\mathbb{E}[Y \mid X] - Y, Y) = \text{Var}(\mathbb{E}[Y \mid X]) - \text{Var}(Y) = -\mathbb{E}[\text{Var}(Y \mid X)].$$

□

This identity explains a source of negative log-residual dependence when important price variation remains unresolved after conditioning on X . It does not imply that every finite-sample model is regressive under every assessor metric. It also does not imply that adding features cannot improve both accuracy and equity.

C Covariance and PRD on the Price Scale

Let $P > 0$ and $r = \hat{P}/P > 0$. The population PRD can be written as

$$\text{PRD} = \frac{\mathbb{E}[r]\mathbb{E}[P]}{\mathbb{E}[rP]}.$$

Then

Proposition 2.

$$\text{sign}(\text{Cov}(r, P)) = \text{sign}(1 - \text{PRD}).$$

Proof. Since $\mathbb{E}[rP] > 0$,

$$1 - \text{PRD} = \frac{\mathbb{E}[rP] - \mathbb{E}[r]\mathbb{E}[P]}{\mathbb{E}[rP]} = \frac{\text{Cov}(r, P)}{\mathbb{E}[rP]}.$$

□

The penalty in the paper acts on $\text{Cov}(\log r, \log P)$, not on $\text{Cov}(r, P)$. Proposition 2 therefore gives directional motivation rather than an equality between the training target and PRD.

D Fixed-Space Regularization Paths: Direct and Surrogate

This appendix isolates the data-fit geometry of the two penalties when the prediction space is held fixed, omitting model-specific complexity penalties in order to expose the correction mechanisms cleanly. The Direct result provides an exact rank-one benchmark for the targeted covariance correction. The Surrogate result provides the corresponding log-price-distance-weighted projection benchmark and makes precise why its path need not remain one-dimensional. These finite-sample projection results are used for mechanism interpretation only; they do not assert that a fully retrained boosting path remains in a fixed prediction space or that the fixed-space formulas describe LightGBM’s finite boosting trajectory exactly.

D.1 Direct penalty: rank-one covariance path

Let $\mathcal{V} \subseteq \mathbb{R}^n$ be a closed linear prediction space, define

$$\langle u, v \rangle_n = \frac{1}{n} u^\top v, \quad \|u\|_n^2 = \langle u, u \rangle_n, \quad c = y - \bar{y}\mathbf{1},$$

and consider

$$f_\rho \in \arg \min_{f \in \mathcal{V}} \left\{ \|f - y\|_n^2 + \frac{\rho}{2} \langle f - y, c \rangle_n^2 \right\}. \quad (39)$$

Let $P_{\mathcal{V}}$ be the orthogonal projection under $\langle \cdot, \cdot \rangle_n$ and set

$$f_0 = P_{\mathcal{V}} y, \quad C_0 = \langle f_0 - y, c \rangle_n, \quad g = P_{\mathcal{V}} c, \quad A = \|g\|_n^2.$$

Proposition 3 (Covariance shrinkage path). *If $A > 0$, then*

$$f_\rho = f_0 - \frac{\rho}{2} C_\rho g, \quad C_\rho = \frac{C_0}{1 + \rho A/2}.$$

Consequently, the slope of residuals on y is multiplied by

$$q(\rho) = \frac{1}{1 + \rho A/2}.$$

Proof. For every $h \in \mathcal{V}$, first-order optimality gives

$$2\langle f_\rho - y, h \rangle_n + \rho C_\rho \langle c, h \rangle_n = 0.$$

Since $f_0 = P_{\mathcal{V}} y$, $\langle f_0 - y, h \rangle_n = 0$. Writing $d_\rho = f_\rho - f_0 \in \mathcal{V}$ and $g = P_{\mathcal{V}} c$ gives

$$2d_\rho + \rho C_\rho g = 0.$$

Taking the inner product with c yields

$$C_\rho = C_0 - \frac{\rho}{2} A C_\rho,$$

which proves the result. □

Corollary 1 (Centered-spread form of the fixed-space Direct path). *If $\mathbf{1} \in \mathcal{V}$, then $g = f_0 - \bar{y}\mathbf{1}$ and*

$$f_\rho = \bar{y}\mathbf{1} + b_\rho(f_0 - \bar{y}\mathbf{1}), \quad b_\rho = 1 - \frac{\rho}{2}C_\rho.$$

Moreover, $C_0 = -\|f_0 - y\|_n^2$. Hence, whenever the unpenalized fit is non-perfect, $C_0 < 0$, $b_\rho > 1$ for $\rho > 0$, and C_ρ approaches zero from below without changing sign at any finite ρ .

Proof. Because $\mathbf{1} \in \mathcal{V}$,

$$g = P_{\mathcal{V}}(y - \bar{y}\mathbf{1}) = f_0 - \bar{y}\mathbf{1}.$$

Substituting this identity into Proposition 3 gives the centered-spread representation. Also,

$$c = (f_0 - \bar{y}\mathbf{1}) + (y - f_0),$$

and the projection residual $y - f_0$ is orthogonal to $\mathcal{V}$, so

$$C_0 = \langle f_0 - y, c \rangle_n = -\|f_0 - y\|_n^2.$$

The remaining statements follow from $C_\rho = C_0/(1 + \rho A/2)$. $\square$

Remark 1 (Mechanical component of the in-sample residual–outcome covariance). Corollary 1 implies that, in this fixed-space squared-error benchmark with an intercept, C_0 is nonpositive even without imposing any equity interpretation: $C_0 = -\|f_0 - y\|_n^2$. The negative training covariance therefore contains a mechanical residual–outcome component induced by fitting the same observed y that is later used to define the log valuation ratio. This identity does not hold mechanically on a new evaluation sample, and it does not imply that observed out-of-sample regressivity is spurious. Rather, it limits the interpretation of the training penalty: $C(f)$ is a controllable first-order mechanism, not a standalone estimator of inequity relative to latent market value. This is why the empirical analysis evaluates the fitted paths chronologically with proxy-aware assessor measures, ratio profiles, and residual-structure diagnostics.

This equivalence is specific to the exact fixed-prediction-space benchmark. It reveals that, within that benchmark, Direct covariance regularization is no richer than a one-parameter centered spread correction. That makes it a mechanism result with an empirical prediction attached, and the prediction is tested rather than deferred: whether an explicit post-hoc correction reproduces the accuracy–equity path of a fully re-trained nonlinear learner is answered for this application in Sections 5.4 and 5.5, where it reproduces much of the Direct first-order frontier over the range Direct attains and does not reproduce the residual-structure behaviour of the Surrogate. It does not imply that the empirical Direct-LightGBM path is one-dimensional: retraining can change tree splits and leaf structure, and the standard-library Direct implementation uses only the diagonal of the covariance Hessian.

The rescaling coefficient that this benchmark implies should be stated in its general form, because the special case is easy to mistake for the rule. On a fitting block T , the coefficient that drives the first-order association of a centered rescaling of a fixed predictor f_0 to zero is

$$b_T^* = \frac{\widehat{\text{Var}}_T(y)}{\widehat{\text{Cov}}_T(f_0, y)},$$

which is the form the frozen post-hoc comparator convention records. The familiar $1/R^2$ expression is the special case in which f_0 is the orthogonal projection of y onto the prediction space, so that $\widehat{\text{Cov}}_T(f_0, y) = \widehat{\text{Var}}_T(f_0)$ and the ratio collapses. It is a theoretical diagnostic confined to that linear/projection benchmark, and it is not the general root for a boosted predictor. Moreover b_T^* is itself an in-sample theory diagnostic rather than an empirical endpoint: the boosted learner used here is substantially less regressive on its own fitting block than out of sample, so the in-sample root understates the rescaling that an out-of-sample first-order correction requires. The frozen convention records it as a diagnostic and does not use it to set any empirical endpoint.

Proposition 4 (Accuracy cost). *Under Proposition 3,*

$$\|f_\rho - y\|_n^2 - \|f_0 - y\|_n^2 = \frac{C_0^2}{A} (1 - q(\rho))^2.$$

Within a fixed prediction space, the targeted covariance falls to first order near $\rho = 0$, while the in-sample squared-error cost is second order. For boosted trees this is a local benchmark only. Retraining can change splits, leaves, and stopping iteration.

D.2 Surrogate penalty: weighted-projection path

Let

$$D = \text{diag}(c_1^2, \dots, c_n^2), \quad W_\rho = I + \rho D, \quad e_0 = f_0 - y,$$

where $f_0 = P_{\mathcal{V}}y$ is the ordinary least-squares projection defined above. The exact fixed-space Surrogate problem corresponding to Eq. (24) is

$$f_\rho^{\text{surr}} = \arg \min_{f \in \mathcal{V}} \{ \|f - y\|_n^2 + \rho \langle f - y, D(f - y) \rangle_n \}. \quad (40)$$

Because W_ρ is positive definite for every $\rho \geq 0$, this problem has a unique solution. Define the weighted inner product

$$\langle u, v \rangle_{n, W_\rho} = \langle u, W_\rho v \rangle_n,$$

let $P_{\mathcal{V}}^{W_\rho}$ denote projection onto $\mathcal{V}$ under this inner product, and define the self-adjoint positive-semidefinite operator

$$T = P_{\mathcal{V}} D|_{\mathcal{V}}$$

together with the baseline tail-weighted residual direction

$$g_{\text{surr}} = P_{\mathcal{V}} D e_0.$$

Proposition 5 (Surrogate weighted-projection path). *For every $\rho \geq 0$,*

$$f_\rho^{\text{surr}} = P_{\mathcal{V}}^{W_\rho} y, \quad (41)$$

and its departure from the unpenalized fit is

$$f_\rho^{\text{surr}} - f_0 = -\rho (I_{\mathcal{V}} + \rho T)^{-1} g_{\text{surr}}. \quad (42)$$

Proof. For every $h \in \mathcal{V}$, first-order optimality for (40) gives

$$\langle f_\rho^{\text{surr}} - y, h \rangle_n + \rho \langle D(f_\rho^{\text{surr}} - y), h \rangle_n = 0,$$

or equivalently

$$\langle f_\rho^{\text{surr}} - y, h \rangle_{n, W_\rho} = 0.$$

This is exactly the weighted-projection characterization in (41). Now write

$$\delta_\rho = f_\rho^{\text{surr}} - f_0 \in \mathcal{V}.$$

Since $f_0 = P_{\mathcal{V}}y$, the ordinary projection residual $e_0 = f_0 - y$ is orthogonal to $\mathcal{V}$. Substituting $f_\rho^{\text{surr}} - y = e_0 + \delta_\rho$ into the first-order condition and projecting the D terms onto $\mathcal{V}$ yields

$$\delta_\rho + \rho T \delta_\rho + \rho g_{\text{surr}} = 0.$$

Because T is positive semidefinite, $I_{\mathcal{V}} + \rho T$ is invertible for $\rho \geq 0$, which proves (42). $\square$

Corollary 2 (Local Surrogate direction and baseline-loss cost). *As $\rho \downarrow 0$,*

$$f_\rho^{\text{surr}} = f_0 - \rho g_{\text{surr}} + O(\rho^2) = f_0 - \rho P_{\mathcal{V}} [c^{\odot 2} \odot e_0] + O(\rho^2). \quad (43)$$

where $\odot$ denotes elementwise multiplication and $c^{\odot 2} = (c_1^2, \dots, c_n^2)^\top$. Moreover,

$$\|f_\rho^{\text{surr}} - y\|_n^2 - \|f_0 - y\|_n^2 = \rho^2 \|g_{\text{surr}}\|_n^2 + O(\rho^3). \quad (44)$$

Proof. The expansion

$$(I_{\mathcal{V}} + \rho T)^{-1} = I_{\mathcal{V}} - \rho T + O(\rho^2)$$

in (42) gives (43). For the second claim, $\delta_\rho = f_\rho^{\text{surr}} - f_0$ belongs to $\mathcal{V}$ and $e_0 = f_0 - y$ is orthogonal to $\mathcal{V}$, so

$$\|e_0 + \delta_\rho\|_n^2 - \|e_0\|_n^2 = \|\delta_\rho\|_n^2.$$

Using $\delta_\rho = -\rho g_{\text{surr}} + O(\rho^2)$ gives (44). $\square$

D.3 Direct–Surrogate comparison in a fixed prediction space

The preceding results expose a structural difference that is not visible from the upper-bound relation alone. For the Direct penalty, Proposition 3 shows that the entire fixed-space path moves along the single direction $P_{\mathcal{V}}c$; when $\mathbf{1} \in \mathcal{V}$, Corollary 1 identifies this direction with the centered fitted-value spread $f_0 - \bar{y}\mathbf{1}$. The Direct correction is therefore rank one in this benchmark. For the Surrogate, the initial direction is instead

$$-g_{\text{surr}} = -P_{\mathcal{V}}[c^{\odot 2} \odot e_0],$$

the learnable component of the baseline residual pattern after weighting observations by squared log-price distance from the sample center. The complete Surrogate path generally changes direction as ρ varies.

To see this explicitly, let $\{u_j\}$ be an orthonormal eigenbasis of T on $\mathcal{V}$, with

$$Tu_j = \lambda_j u_j, \quad \lambda_j \geq 0,$$

and write

$$g_{\text{surr}} = \sum_j a_j u_j.$$

Then Proposition 5 gives

$$f_{\rho}^{\text{surr}} - f_0 = - \sum_j \frac{\rho}{1 + \rho \lambda_j} a_j u_j. \quad (45)$$

Thus, unless the nonzero components of g_{surr} lie in a single eigenspace of T , the relative contribution of the learnable correction modes changes with ρ . This contrasts with the one-direction Direct path. It also makes clear that the Surrogate does not inherit the Direct formula $C_{\rho} = C_0/(1 + \rho A/2)$: the Surrogate minimizes a log-price-distance-weighted prediction loss rather than the signed covariance itself, so monotone covariance shrinkage is not guaranteed by the fixed-space theory. Likewise, the spectral representation does not imply that stronger Surrogate regularization must generate any particular nonlinear ratio shape; the empirical shape and Δ_{NL} results remain separate diagnostics.

Taken together, the fixed-space results clarify the distinct mechanisms of the two methods. Direct regularization applies a rank-one correction aligned with the projected centered-price direction and, with an intercept in the prediction space, becomes an exact centered spread rescaling. Surrogate regularization changes the projection geometry through the diagonal weights $1 + \rho c_i^2$: locally it repairs the learnable tail-weighted residual pattern, and globally it can combine multiple correction modes. In both cases the baseline squared-error cost is second order near $\rho = 0$, but only the Direct objective guarantees monotone shrinkage of the targeted covariance in this benchmark. The comparison therefore supports the paper’s interpretation of the Surrogate as a related but distinct objective rather than as a purely computational approximation to Direct.

E Linear and Boosting Implementations

E.1 Linear Model

Let $X \in \mathbb{R}^{n \times p}$ and $e = X\theta - y$. With

$$c = y - \bar{y}\mathbf{1}, \quad a = X^{\top}c, \quad b = y^{\top}c,$$

the direct penalized estimator under loss $\|X\theta - y\|_2^2/n$ satisfies

$$\left(\frac{1}{n} X^{\top} X + \frac{\rho}{2n^2} aa^{\top} \right) \theta = \frac{1}{n} X^{\top} y + \frac{\rho}{2n^2} ba.$$

This is a rank-one modification of the normal equations.

For the surrogate, let $D = \text{diag}(c_1^2, \dots, c_n^2)$. Then

$$\left(\frac{1}{n} X^{\top} X + \frac{\rho}{n} X^{\top} D X \right) \theta = \frac{1}{n} X^{\top} y + \frac{\rho}{n} X^{\top} D y.$$

E.2 LightGBM Custom-Objective Initialization and Scaling

For the canonical custom objectives, let $b = \bar{y}$ denote the mean log sale price in the current training sample. We fit the custom-objective model to centered labels $\tilde{y}_i = y_i - b$ with zero initial score and add b back to predictions. Equivalently, the fitted model predicts on the original log-price scale as $f(x) = b + \hat{f}(x)$. This construction is designed to match the native squared-error LightGBM starting prediction while leaving $e_i = f(x_i) - y_i$ and $c_i = y_i - \bar{y}$ unchanged.

The corresponding native/custom zero-penalty audit is reported in the next subsection.

The mathematical objectives in the main text are written as sample averages. In the LightGBM custom-objective implementation we multiply the complete objective by $n/2$, a common positive factor that does not change its minimizer. This yields a native squared-error baseline gradient $g_i = e_i$ and Hessian $h_i = 1$ at $\rho = 0$. For the direct objective, the supplied gradient and diagonal Hessian are

$$g_i^{\text{cov}} = e_i + \frac{\rho}{2} z c_i, \quad h_i^{\text{cov,diag}} = 1 + \frac{\rho}{2n} c_i^2, \quad z = \frac{1}{n} \sum_{i=1}^n e_i c_i.$$

For the surrogate, the supplied derivatives are

$$g_i^{\text{surr}} = e_i(1 + \rho c_i^2), \quad h_i^{\text{surr}} = 1 + \rho c_i^2.$$

Thus the direct implementation uses the exact covariance gradient and only the diagonal of its dense covariance curvature, whereas the surrogate derivatives are exactly observation-additive. The $\rho = 0$ custom-objective path is retained as an implementation control.

To assess whether the omitted Direct curvature could be treated as negligible over the evaluated configurations, we examined two implementation diagnostics. The maximum curvature ratio at the display anchors is 40.04, compared with the diagnostic threshold of 1.2 for $\rho \geq 1$, and the maximum penalty contribution M over all fitting blocks and anchors is 0.0118, compared with a threshold of 0.01. Neither diagnostic supports treating the omitted curvature as negligible. At the same time, these diagnostics do not establish that the omission materially changes the fitted regularization path; the main text therefore makes neither claim.

E.3 Zero-Penalty Implementation Audit

Before attributing within-family changes to positive regularization, we compare ordinary LightGBM with the Direct and Surrogate implementations at $\rho = 0$. The two custom-objective controls coincide with each other in the reported metrics, but the currently frozen artifacts are not numerically identical to ordinary LightGBM. Because the supplied zero-penalty gradient and Hessian match squared-error derivatives algebraically, this audit identifies a native-versus-custom execution-path difference rather than a positive-penalty objective effect. The paragraphs after the table report what the completed audit then established, and the boundary of that claim: the evidence isolates the setting responsible and the boosting iteration at which the two paths separate, but it does not directly instrument LightGBM’s internal random-number stream, so no claim is made about library internals beyond what the tree, split and source evidence supports.

Table 9: Implementation-control comparison at $\rho = 0$. Prediction differences are log-price deviations from ordinary LightGBM, aligned on sale identifiers and computed independently on the held-out and 2025 samples.

Sample	Model	R_P^2	$\text{RMSE}_{\log P}$	$\beta_{\log}$	Mean $ \Delta \hat{y} $	Max $ \Delta \hat{y} $
Held-out	Ordinary LightGBM	0.894	0.289	-0.150	—	—
Held-out	Direct, $\rho = 0$	0.893	0.289	-0.147	3.24e-02	3.09e-01
Held-out	Surrogate, $\rho = 0$	0.893	0.289	-0.147	3.24e-02	3.09e-01
2025 forward	Ordinary LightGBM	0.904	0.278	-0.164	—	—
2025 forward	Direct, $\rho = 0$	0.904	0.279	-0.161	3.06e-02	2.83e-01
2025 forward	Surrogate, $\rho = 0$	0.904	0.279	-0.161	3.06e-02	2.83e-01

The native-versus-custom decomposition, as measured. The native-versus-custom parity experiment has been carried out, and it resolves in the negative. It compares the three cells at three learner capacities — a small and an intermediate diagnostic capacity, and the canonical capacity that produced the reported paths — on both out-of-time samples. Ordinary LightGBM and the centered-label native L2 control agree to numerical tolerance at the two reduced capacities: at the small capacity their held-out mean absolute log difference is 7.69e-09 with a maximum of 2.58e-08, and the intermediate capacity is of the same order. The centered-label native L2 control and the custom-objective $\rho = 0$ origin do *not* agree at any tested capacity: the same held-out comparison gives a mean absolute log difference of 1.76e-02 with a maximum of 1.40e-01 at the small capacity, and 3.26e-02 with a maximum of 2.97e-01 at the canonical capacity. So the label-centering step is not what separates the two implementations. At the canonical capacity a second and separate effect — the single-precision representation of the centered labels, amplified by capacity — also enters the ordinary-versus-centered comparison, which is why the clean label-representation decomposition is legible only at reduced capacity.

Two further results fix the interpretation. Repeating the whole ladder with LightGBM’s determinism controls engaged reproduces the unpinned ladder cell for cell, with the same centered-label-to-custom canonical-capacity held-out maximum of 2.97e-01 under both, so pinning does not remove the difference and run-to-run numerical variation cannot account for it; the same-host replicate check puts that numerical floor at exactly zero. Tracing single-tree fits then isolates one setting. With the frozen `colsample_bytree` of 0.541 the two learners already differ in the structure of the first tree, and pinning the histogram path changes nothing; setting `colsample_bytree` to 1 makes them bitwise identical. The accepted characterization is therefore a deterministic built-in-versus-custom feature-subsampling execution-path divergence under the canonical feature-subsampling setting. It is not evidence of an error in the covariance objective: given the same feature subsets, the custom $\rho = 0$ objective reproduces native squared-error learning exactly. Because `colsample_bytree` was tuned as part of the canonical configuration, setting it to 1 specifies a different learner rather than reproducing the frozen experiment, so it is not an available repair. Table 9 is accordingly reported as measured and is not regenerated, and the workflow-benchmark comparisons in the Results are labelled as such rather than read as penalty effects.

Normalization and scaling, as audited. The executed code has been audited against the normalization in Eqs. (4), (17) and (22) and against the $n/2$ LightGBM scaling stated above. The audit covers all nine fitting blocks — the seven chronological training blocks, the development pool, and the production block used for the forward year — and records the same three facts in each. The $n/2$ factor is applied analytically, so the supplied derivatives reduce to $g_i = e_i$ and $h_i = 1$ at $\rho = 0$. The supplied Direct diagonal Hessian is $1 + (\rho/2n)c_i^2$, against an exact scaled Hessian $I + (\rho/2n)cc^\top$ whose off-diagonal rank-one term the observation-wise interface cannot carry. The supplied Surrogate Hessian is $1 + \rho c_i^2$. The executed LightGBM parameter vector is pinned by the configuration hash `8f0f2acd83118de782604b5ca7143acfb2af3fd186ea9376588f9bcf560585b`, re-verified against the frozen record. The source state that produced those artifacts is bounded rather than reconstructed. All three generating commits are ancestors of the frozen provenance head, and the objective, model-factory, split and metric files are byte-identical at every one of them; the two executed-path files that did drift are archived as per-file patches and adjudicated behaviour-neutral for prediction reproduction. All three generating trees were nonetheless dirty, and only a diff hash survives for two of them and none for the third, so the uncommitted state cannot be reconstructed from what was recorded. The frozen provenance discloses that residual as an irreducible limitation rather than resolving it. No factor-of-two or sample-size convention is absorbed into a relabelled ρ : the ρ values indexing the reported paths are the values the executed objective received.

E.4 Direct Penalty in Boosting

For current predictions f , define $e = f - y$ and

$$z = \frac{1}{n} e^\top c.$$

Under squared loss, the direct objective has

$$g_i = \frac{2e_i}{n} + \frac{\rho}{n} z c_i,$$

and exact Hessian

$$\nabla_f^2 = \frac{2}{n}I + \frac{\rho}{n^2}cc^\top.$$

The diagonal approximation uses

$$h_i^{\text{diag}} = \frac{2}{n} + \frac{\rho}{n^2}c_i^2.$$

For a fixed tree with leaves $I_1, \dots, I_J$, let

$$b_j = \sum_{i \in I_j} e_i, \quad s_j = \sum_{i \in I_j} c_i, \quad n_j = |I_j|.$$

With leaf-weight vector w , ridge parameter λ , and $s = (s_1, \dots, s_J)^\top$, the exact local quadratic system is

$$A_T w^* = -g_T, \quad g_{T,j} = \frac{2b_j}{n} + \frac{\rho z s_j}{n},$$

$$A_T = \text{diag}\left(\frac{2n_1}{n} + \lambda, \dots, \frac{2n_J}{n} + \lambda\right) + \frac{\rho}{n^2}ss^\top.$$

Thus $w^* = -A_T^{-1}g_T$ when A_T is nonsingular. This solver retains the cross-observation curvature but requires a custom tree-construction procedure; it is not available through the usual observation-wise API.

E.5 Surrogate in Boosting

For

$$\Psi^{\text{surr}}(f) = \frac{1}{n} \sum_i e_i^2 c_i^2,$$

the derivatives are observation-wise:

$$g_i = \frac{2e_i}{n} + \frac{2\rho}{n}e_i c_i^2, \quad h_i = \frac{2}{n} + \frac{2\rho}{n}c_i^2.$$

This is the canonical surrogate implementation evaluated in the CCAO regularization path.

F Primary CCAO Regularization-Path Details

This appendix contains the complete diagnostics underlying the compact main-text regularization-path presentation. It is organized around the final augmented paths and not around a selected configuration; the compact nominal decade-spaced anchors are presentation-only snapshots of those frozen paths.

F.1 Complete Regularization-Path Results

The complete regularization path is held in the frozen combined path table, which carries every family- ρ configuration together with the seven fold values, equal-weight CV mean and standard deviation, held-out results, and 2025 forward results for the Section 2.3 evaluation metrics. That table is an output of the executed run root rather than a released file: the frozen evidence manifest identifies it by path, byte length and content hash, and it is not redistributed with this paper. Appendix H sets out which derived artifacts are tracked and which are hash-identified only. The compact main-text presentation is split in two. Table 6 carries prediction, PRD and MKI at the two positive anchors that are directly reconstructible from retained fitted artifacts, and Table 7 carries PRB, from the adopted Standard on Ratio Studies, and VEI, an Exposure-Draft diagnostic and therefore proposed rather than adopted, at all five nominal decade-spaced anchors $\{0.01, 0.1, 1, 10, 100\}$ mapped to the nearest tested grid points. Both show the ordinary-LightGBM workflow benchmark and the custom-objective zero-penalty origin as explicit rows; the wider zero-penalty implementation audit stays in the implementation appendix. No path row is omitted from the machine-readable results because of its observed performance. For completeness, the complementary

Table 10: Complementary baseline level for *Ordinary LightGBM (standard raw-label native)*, the assessor-facing workflow benchmark: valuation level, horizontal uniformity, and mechanism/residual structure. A level statement about one model class, not a comparison between model classes.

Measure	Held-out evaluation	2025 forward evaluation
Valuation level		
Median ratio m_S	0.929	0.950
Mean ratio $\bar{r}_S$	0.989	1.015
Weighted mean $\bar{r}_{W,S}$	0.924	0.941
Horizontal uniformity		
COD	21.6%	21.3%
COV	39.7%	37.2%
Mechanism and residual structure		
$\beta_{\log}$	-0.150	-0.164
Δ_{NL}	0.119	0.121
$dCor(e, y)$	0.387	0.422

Notes. Every entry reports the same unpenalized workflow benchmark, which carries all eight complementary measures on both evaluations. The earlier township-level linear-regression workflow is discussed only as historical context and is neither re-estimated nor reported quantitatively in this study. Median, mean, weighted-mean ratio, COD, and COV are assessor-facing valuation-level or horizontal-uniformity diagnostics defined in Section 2.3. The first-order mechanism slope $\beta_{\log}$, correlation-ratio nonlinearity gap Δ_{NL} , and distance correlation are defined in Eqs. (11), (12), and (38); the last two carry no assessor-standard reference range and are residual-structure diagnostics rather than equity measures. One reading matters for the motivation and is stated rather than left out: the workflow benchmark’s COD is 21.6% held-out and 21.3% in 2025, both above the [5, 15] range the adopted Standard on Ratio Studies gives for single-family residential property (Table 1), and COD rises further along the penalty paths (Appendix Table 11). No compliance claim is implied anywhere in this paper. This range comparison is a descriptive interpretation of the reported benchmark values rather than a formal inferential finding. COD and COV are reported in percent; COV is stored as a fraction in the retained artifact and converted to percent for display.

baseline diagnostics and the complementary representative- ρ diagnostics are reported here rather than in the main empirical narrative.

The frozen combined path table carries every family- ρ configuration for the Section 2.3 metrics, together with the fold values, the equal-weight CV mean and standard deviation, and the held-out and 2025 forward results; as Appendix H records, it is hash-identified in the frozen evidence manifest rather than redistributed. Chronological-CV Δ_{NL} was reconstructed fold by fold from the already-frozen validation predictions using the same fixed cross-fitted estimator specified in Appendix A; those CV values did not define the ρ grid or any model-selection rule. Split-specific $\rho = 0$ prediction-parity audits belong to the same result root.

F.2 Prediction, Valuation-Level, and Horizontal-Uniformity Paths

The ρ -evolution figures below trace the frozen tested grid descriptively. They mark no penalty interval, and this paper does not select, recommend, or certify a penalty strength.

F.3 Exact Rolling-Origin Fold Structure

Table 12 records the exact cumulative windows and sample sizes underlying the seven-fold chronological design described in Section 4.2. In every fold, the chronologically newest 10% of the cumulative window is used for validation; because that rule is identical across folds, it is stated once here rather than repeated as a table column.

F.4 Chronological Fold Stability

Three alternative temporal designs were evaluated against the primary design without refitting it. A strict-date design moves every boundary-date sale to the evaluation side, so that training strictly precedes evaluation at each origin; an oracle overlap-removal diagnostic deletes from each training block every parcel that reappears in the paired evaluation block; and an unseen-parcel view restricts the retained predictions to evaluation rows whose parcel never appears in the corresponding training block. The alternative designs produced one initial discrepancy in the sign and family ordering of the first-order slope. Targeted local

Table 11: Complementary diagnostics at the two directly reconstructible positive- ρ anchors: valuation level, horizontal uniformity, and the first-order mechanism slope, with both reference cells shown as explicit rows. Boldface marks movement toward the measure’s own stated direction relative to *Ordinary LightGBM (standard raw-label native)*, the assessor-facing workflow benchmark; a dagger ($\dagger$) marks movement in that direction relative to the *Custom-objective $\rho = 0$ origin*, the primary within-path penalty-isolating reference. The two reference rows carry no markers. Neither device denotes statistical significance, standards compliance, model selection, or a preferred penalty strength: no penalty strength is selected anywhere in this paper.

Method	ρ	Median r	Mean r	W. mean r	COD	COV	$\beta_{\log}$
<i>Panel A: Held-out evaluation</i>							
Reference cells							
Ordinary LightGBM (standard raw-label native)	–	0.929	0.989	0.924	21.6%	39.7%	-0.150
Custom-objective $\rho = 0$ origin	0	0.928	0.986	0.923	21.6%	39.6%	-0.147
Direct penalty							
Direct	≈ 0.954095	0.936$\dagger$	0.993$\dagger$	0.936$\dagger$	21.6%	39.9%	-0.134$\dagger$
Direct	100	0.968$\dagger$	1.022	0.993$\dagger$	24.5%	41.5%	-0.079$\dagger$
Surrogate penalty							
Surrogate	≈ 0.954095	0.929$\dagger$	0.975	0.930$\dagger$	21.4%$\dagger$	38.2%$\dagger$	-0.102$\dagger$
Surrogate	100	0.923	0.937	0.928$\dagger$	23.5%	40.3%	-0.001$\dagger$
<i>Panel B: 2025 forward evaluation</i>							
Reference cells							
Ordinary LightGBM (standard raw-label native)	–	0.950	1.015	0.941	21.3%	37.2%	-0.164
Custom-objective $\rho = 0$ origin	0	0.946	1.009	0.936	21.2%	37.1%	-0.161
Direct penalty							
Direct	≈ 0.954095	0.957$\dagger$	1.019	0.954$\dagger$	21.2%$\dagger$	37.2%	-0.148$\dagger$
Direct	100	0.992$\dagger$	1.049	1.012$\dagger$	23.3%	40.3%	-0.093$\dagger$
Surrogate penalty							
Surrogate	≈ 0.954095	0.950 $\dagger$	1.003$\dagger$	0.946$\dagger$	20.9%$\dagger$	36.1%$\dagger$	-0.120$\dagger$
Surrogate	100	0.948 $\dagger$	0.964	0.945$\dagger$	22.7%	36.8%$\dagger$	-0.015$\dagger$

Notes. Median, mean and weighted-mean ratio, COD and COV are the assessor-facing valuation-level and horizontal-uniformity diagnostics defined in Section 2.3; the first-order mechanism slope $\beta_{\log}$ is defined in Eq. (11). Reference-cell values are taken from the retained zero-penalty control and the two positive- ρ rows from directly reconstructed fitted artifacts. This is why only these two positive anchors appear: the other three nominal display anchors are not directly reconstructible from retained fitted artifacts and are omitted rather than shown blank. The correlation-ratio nonlinearity gap Δ_{NL} and the distance correlation $\text{dCor}(e, y)$ are deliberately *not* columns of this table. The retained fitted artifacts carry neither measure at positive ρ , so no positive- ρ value for either is reported here; the ordinary-LightGBM values are reported in Appendix Table 10 instead. The three valuation-level ratios are compared by distance from one. COD and COV have no universal ideal, and unusually low dispersion can itself warrant diagnostic review, so a marker on either means only that dispersion moved lower, not that the configuration is better; the adopted 2013 COD range for single-family residential is [5, 15], and every COD entry in this table lies above it. A marker on $\beta_{\log}$ means the first-order slope moved toward zero, which is a mechanism direction and not an assessor-standard equity target. Comparisons are evaluated on the unrounded canonical values, so two cells that appear equal after display rounding can still be marked differently. COD and COV are reported in percent.

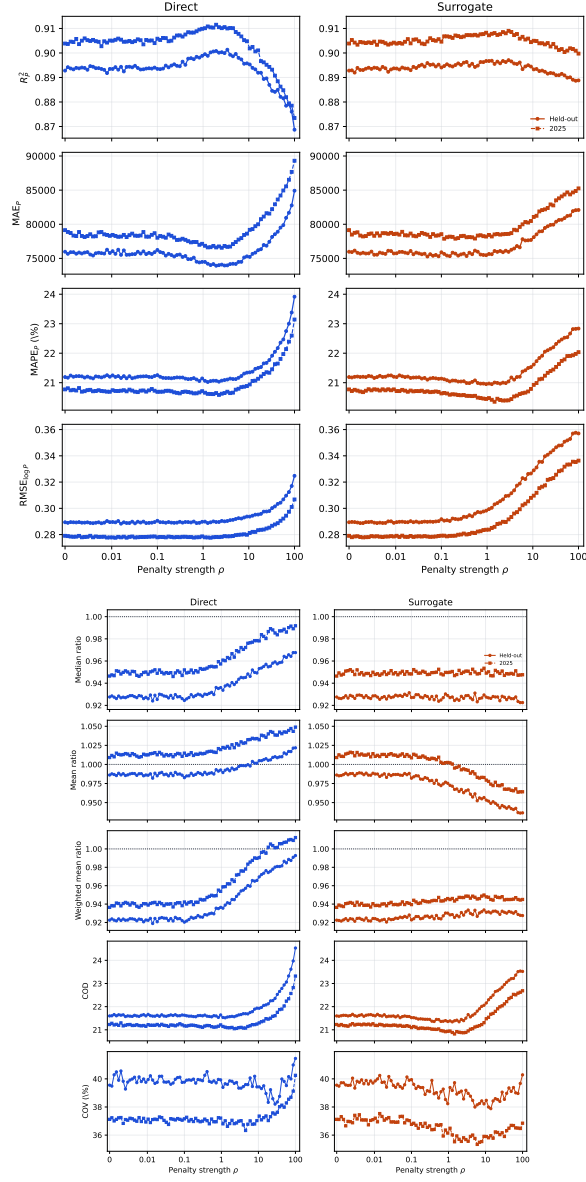


Figure 5: Predictive-metric and valuation-level/uniformity paths along the Direct and Surrogate grids, with held-out and 2025 evaluations overlaid. Valuation-level panels include the ratio = 1 reference. No penalty strength is selected in this paper.

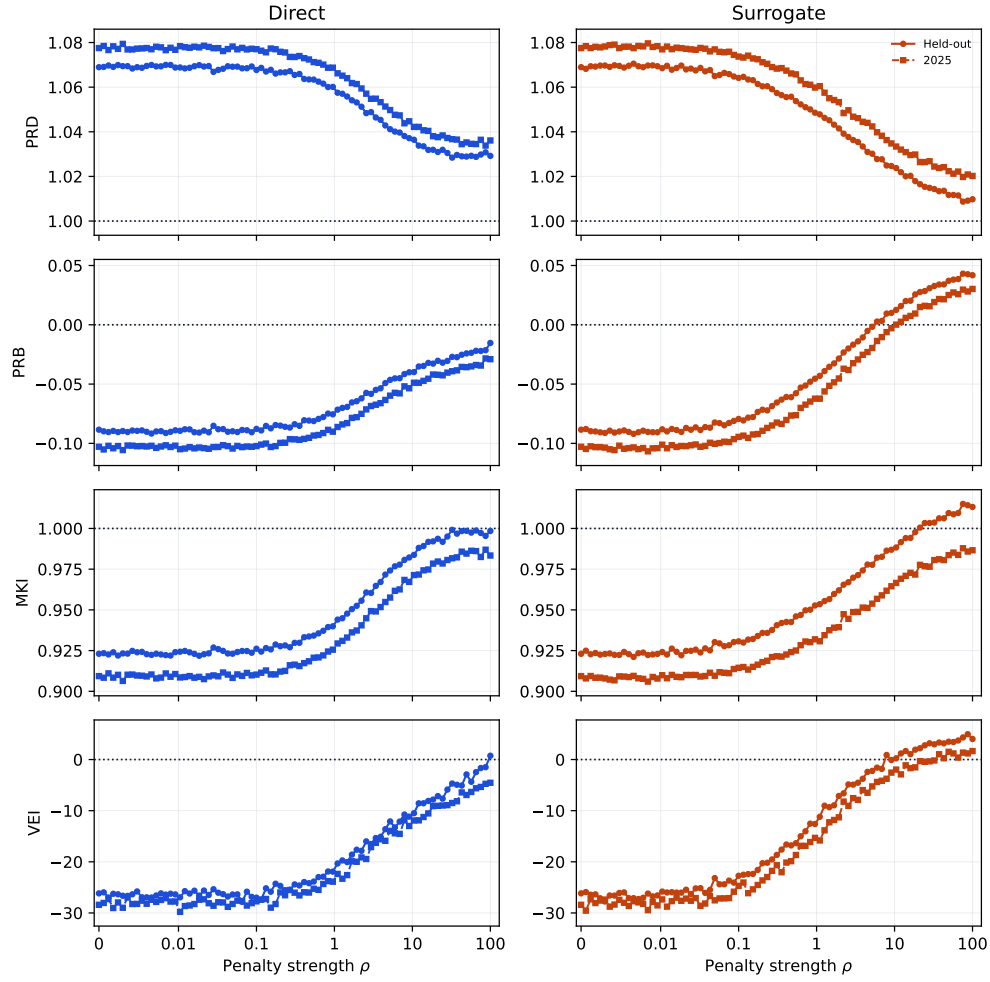


Figure 6: Vertical-equity metric paths versus ρ for Direct and Surrogate. Reference lines mark $\text{PRD} = 1$, $\text{PRB} = 0$, $\text{MKI} = 1$, and $\text{VEI} = 0$. No penalty strength is selected in this paper.

Table 12: Rolling-origin validation structure for the final CCAO development sample. In every fold, the newest 10% of observations in the cumulative window form the validation block. The fold-6 and fold-7 validation blocks overlap; accordingly, fold-level statistics computed across these rows are descriptive chronological-window summaries rather than independent replications, and they support no IID standard errors. See Section 4.2 for the development coordinates and the one-sale-one-vote D3 sensitivity.

Fold	Cumulative window	n_{train}	n_{val}
1	2016-01-01–2017-04-01	46,888	5,209
2	2016-01-01–2018-07-01	100,776	11,197
3	2016-01-01–2019-10-01	151,187	16,798
4	2016-01-01–2021-01-01	200,908	22,323
5	2016-01-01–2022-04-01	252,486	28,053
6	2016-01-01–2023-07-01	298,022	33,113
7	2016-01-01–2023-11-09	310,147	34,460

refinement around the affected region showed that the discrepancy arose from near-zero values and near-ties and did not persist once the local grid was refined. We therefore found no material contradiction to the conclusions under the primary temporal design. The comparison used the same separation tolerance, $\tau = 0.002$, that had already been fixed for the matched-correction analysis before these robustness outcomes were examined.

These designs sharpen rather than soften the reading. Under the strict-date design the first-order path keeps its shape, sign, and family ordering wherever the two families are separated by more than that tolerance, and the 2025 forward numbers are unchanged because that boundary was already date-separated—a property of the design rather than independent confirmation of portability. The oracle diagnostic could not be implemented prospectively, since constructing its training blocks requires knowing which parcels appear in a future evaluation block; it is therefore reported as a diagnostic and not as a proposed sample-construction rule. It leaves the within-path conclusions in place, and it makes the uncorrected origin measurably more regressive, which strengthens rather than weakens the motivation for the correction. The unseen-parcel view refits nothing and changes the denominators, so it is reported on its own base and is never differenced against the frozen path; on it the Surrogate continues to attain substantially more first-order correction than the Direct family at strong penalties. Direct and Surrogate are accordingly not interchangeable, and their attainable spans and temporal behaviour are reported separately rather than averaged together.

The reconstructed CV Δ_{NL} paths in Figure 10 are a retrospective diagnostic. No minimum value and no grid location is quoted from them, because the reconstructed CV path values are not part of the frozen numeric evidence, and no location on those paths is a selected operating point.

F.5 Centered-Spread Post-Hoc Comparator Detail

Construction and centering sensitivity. The comparator rescales a cached prediction array about the mean log *target* of its own fitting block,

$$f_b(x) = \bar{y}_T + b(f_0(x) - \bar{y}_T),$$

where $\bar{y}_T$ is the fold- k training mean for fold k , the development-pool mean for the held-out block, and the 2016–2024 production mean for the 2025 forward block. This is the theorem-matched form, because Corollary 1 writes the fixed-space Direct path in exactly that shape, and it is the only map used for the full path. At $b = 1$ the map returns the cached array unchanged, so each post-hoc path starts at its own reference cell rather than at a re-evaluation of it. No model is refit anywhere in this comparator: the inputs are the frozen prediction arrays, hash-verified before use. The frozen grid holds 124 base values, giving a path table of 2,480 rows.

A second centering — about the fitting block’s mean *prediction* $\bar{f}_{0,T}$ instead of $\bar{y}_T$ — is a named sensitivity rather than an alternative primary map, and the two are never merged. At the same b they differ by $b(\bar{f}_{0,T} - \bar{y}_T)$, a constant in x , so $\beta_{\log}$, the residual–price covariance, COD, COV and distance correlation

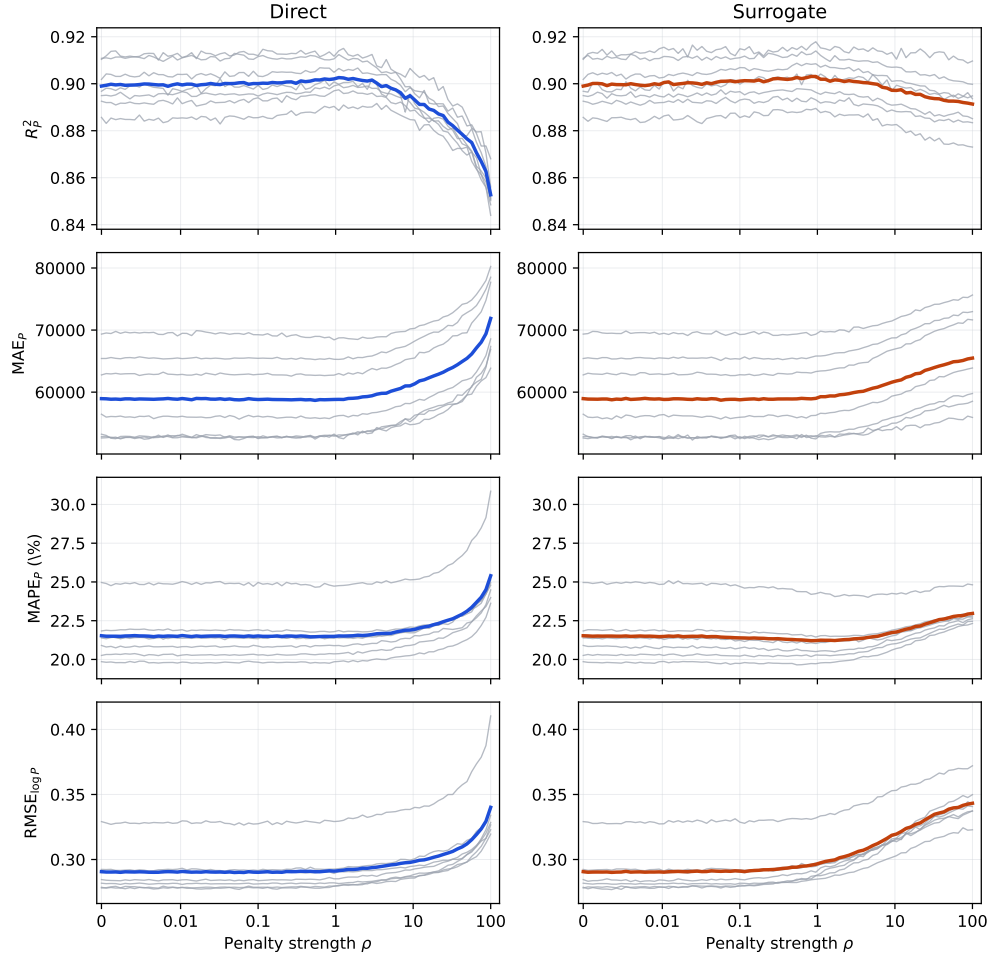


Figure 7: Chronological-fold predictive-metric paths (thin gray) and equal-weight CV means (thick). The seven folds are overlapping cumulative-window constructions rather than independent replications, so their spread is descriptive and supports no standard error. No penalty strength is selected in this paper.

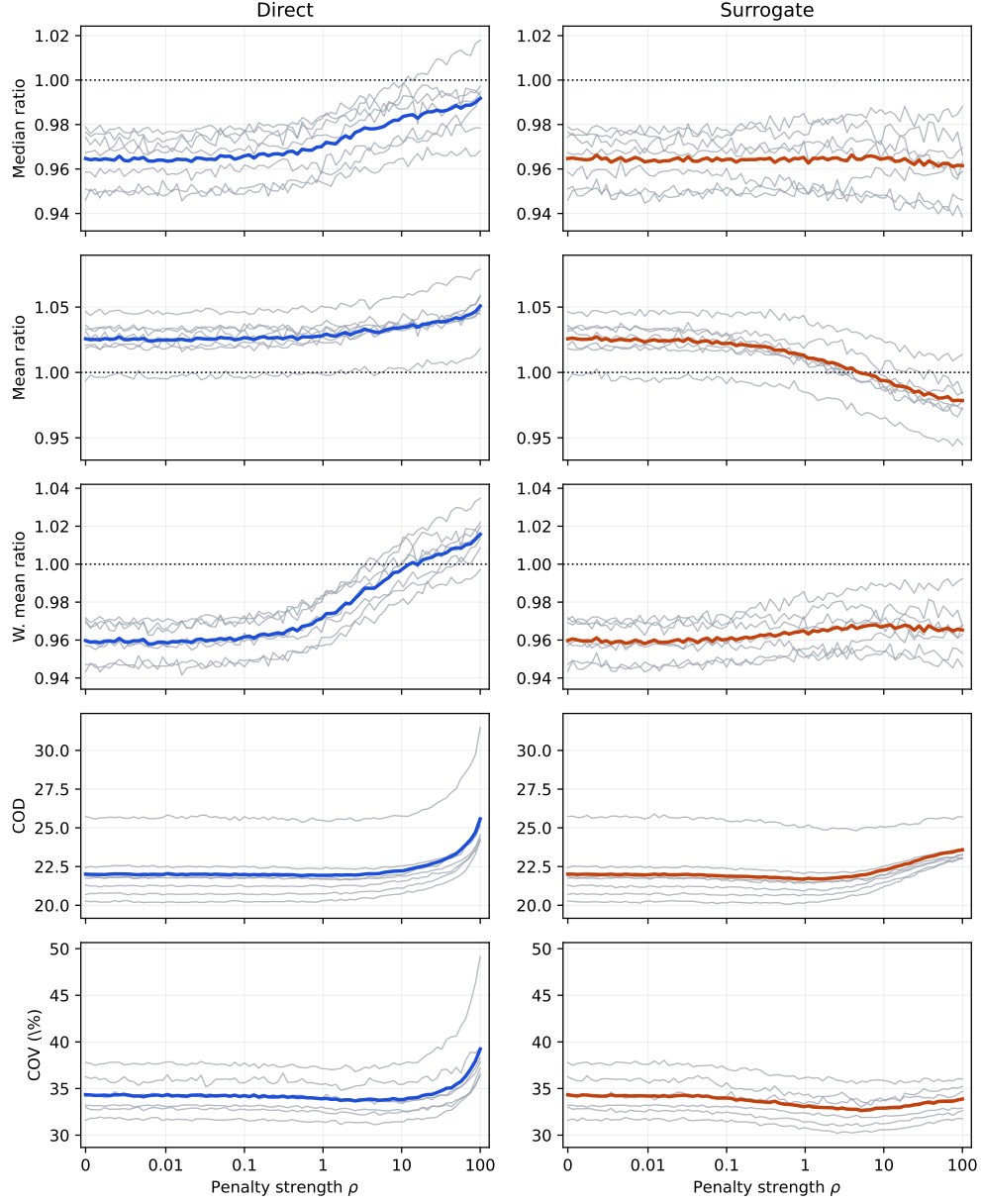


Figure 8: Chronological-fold valuation-level and uniformity paths (thin gray) and equal-weight CV means (thick). Valuation-level panels include the ratio = 1 reference. The seven folds are overlapping cumulative-window constructions rather than independent replications, so their spread is descriptive and supports no standard error. No penalty strength is selected in this paper.

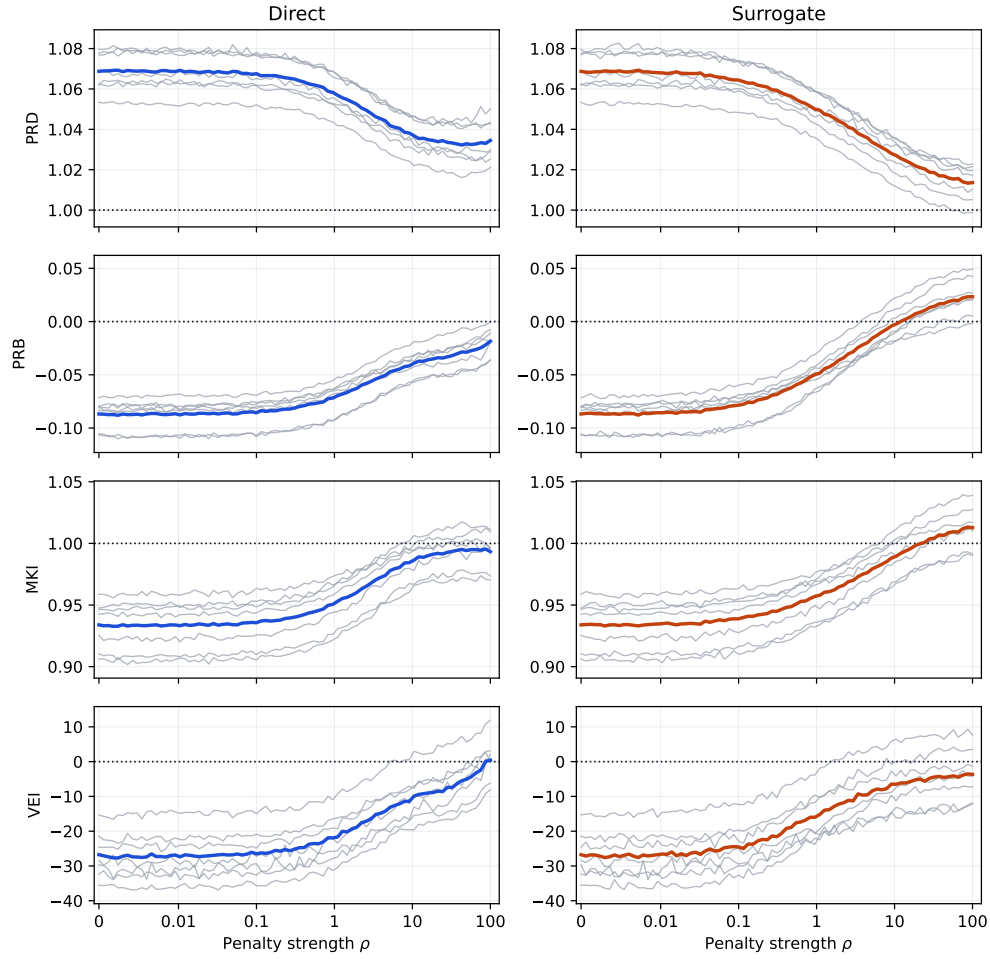


Figure 9: Chronological-fold vertical-equity paths (thin gray) and equal-weight CV means (thick). Reference lines mark $\text{PRD} = 1$, $\text{PRB} = 0$, $\text{MKI} = 1$, and $\text{VEI} = 0$. The seven folds are overlapping cumulative-window constructions rather than independent replications, so their spread is descriptive and supports no standard error. No penalty strength is selected in this paper.

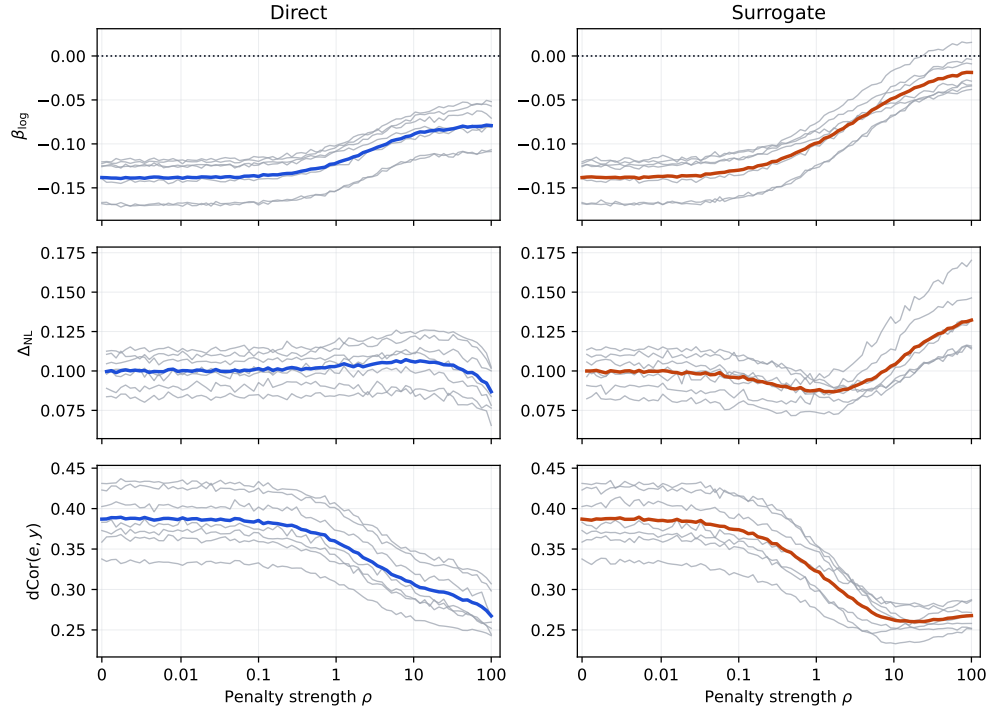


Figure 10: Chronological-fold mechanism paths for $\beta_{\log}$, Δ_{NL} , and distance correlation (thin gray) and equal-weight CV means (thick). The $\beta_{\log} = 0$ line is a first-order neutrality reference; Δ_{NL} and dCor are not given forced zero lines. CV Δ_{NL} was reconstructed from frozen validation predictions with the same fixed estimator used out of time; it is a retrospective diagnostic and does not select ρ . The seven folds are overlapping cumulative-window constructions rather than independent replications, so their spread is descriptive and supports no standard error. No penalty strength is selected in this paper.

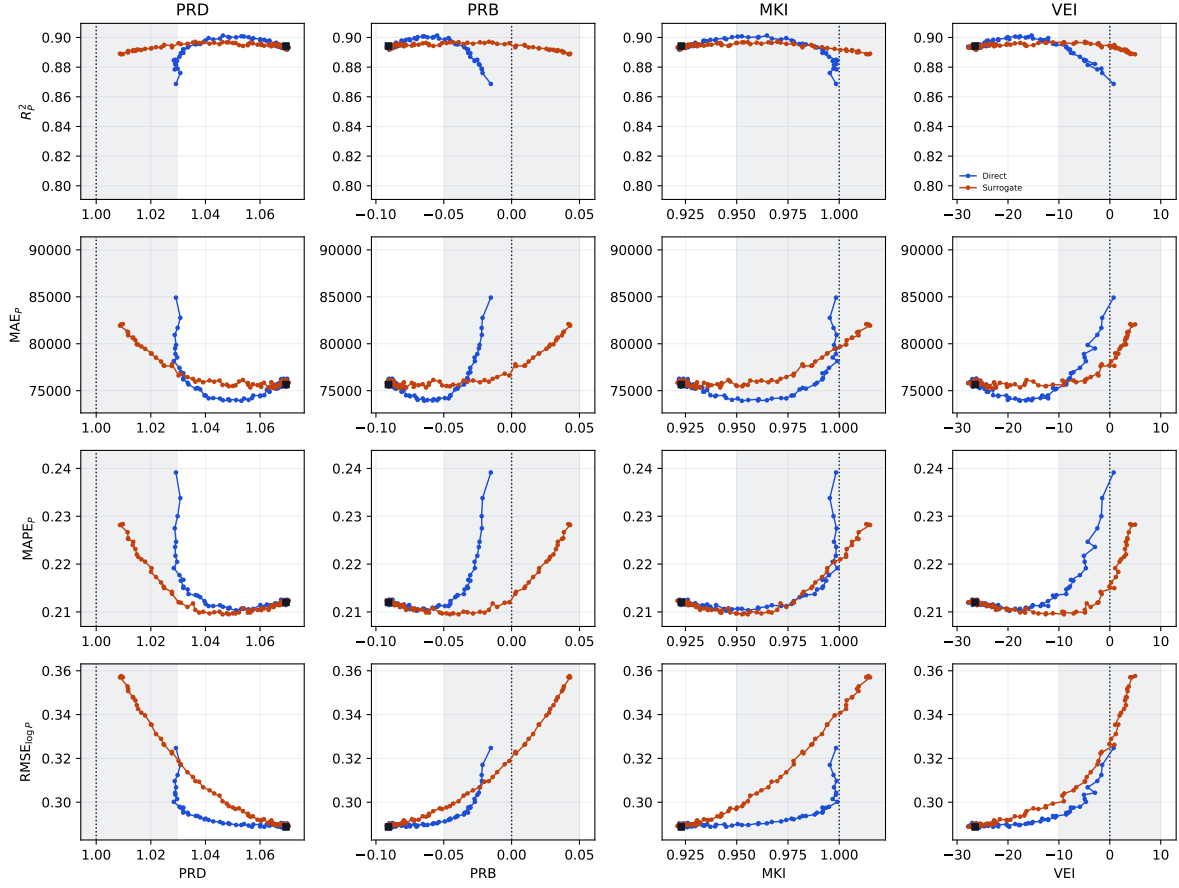


Figure 11: Held-out assessor-facing diagnostics (horizontal) versus predictive metrics (vertical) along the Direct and Surrogate paths. Dotted vertical lines mark metric neutrality where applicable, and shaded regions are assessor-facing guidance ranges rather than compliance claims.

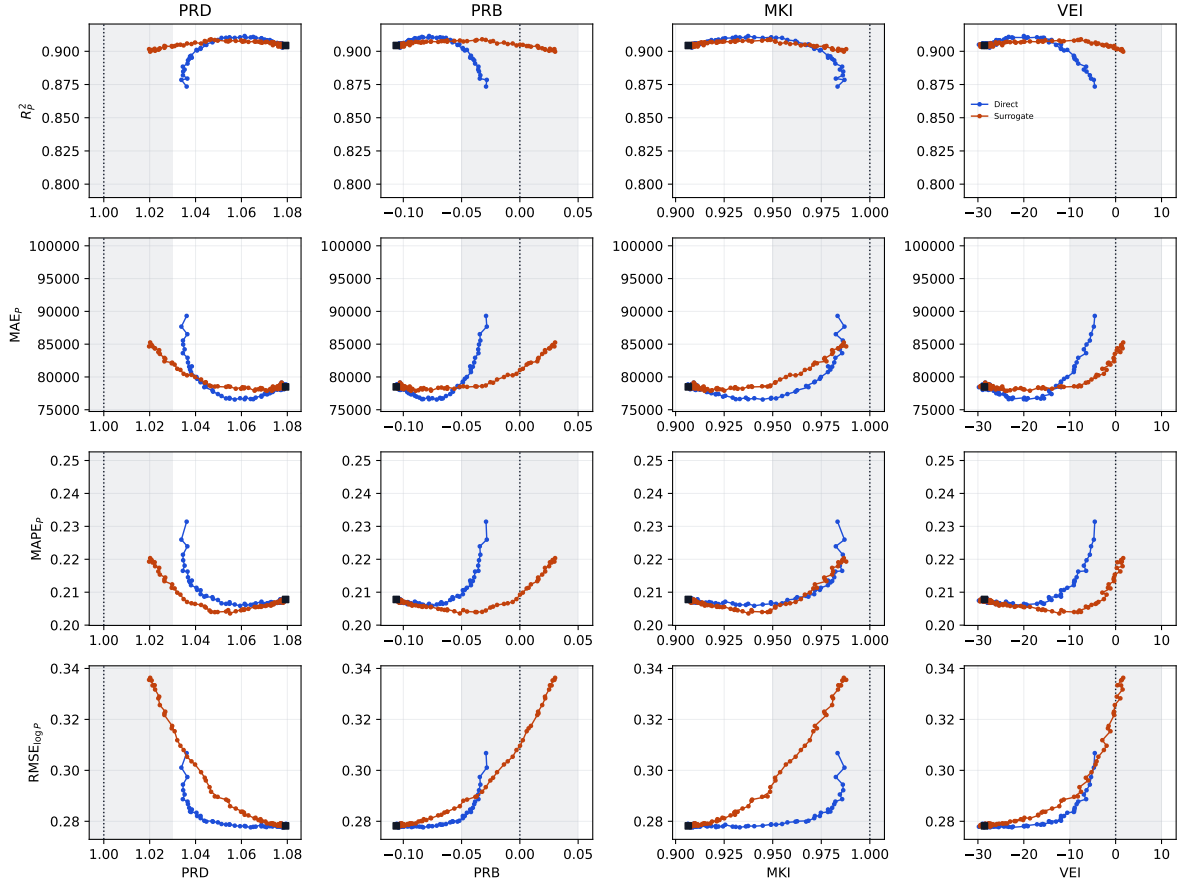


Figure 12: 2025 assessor-facing diagnostics (horizontal) versus predictive metrics (vertical) along the Direct and Surrogate paths. Dotted vertical lines mark metric neutrality where applicable, and shaded regions are assessor-facing guidance ranges rather than compliance claims.

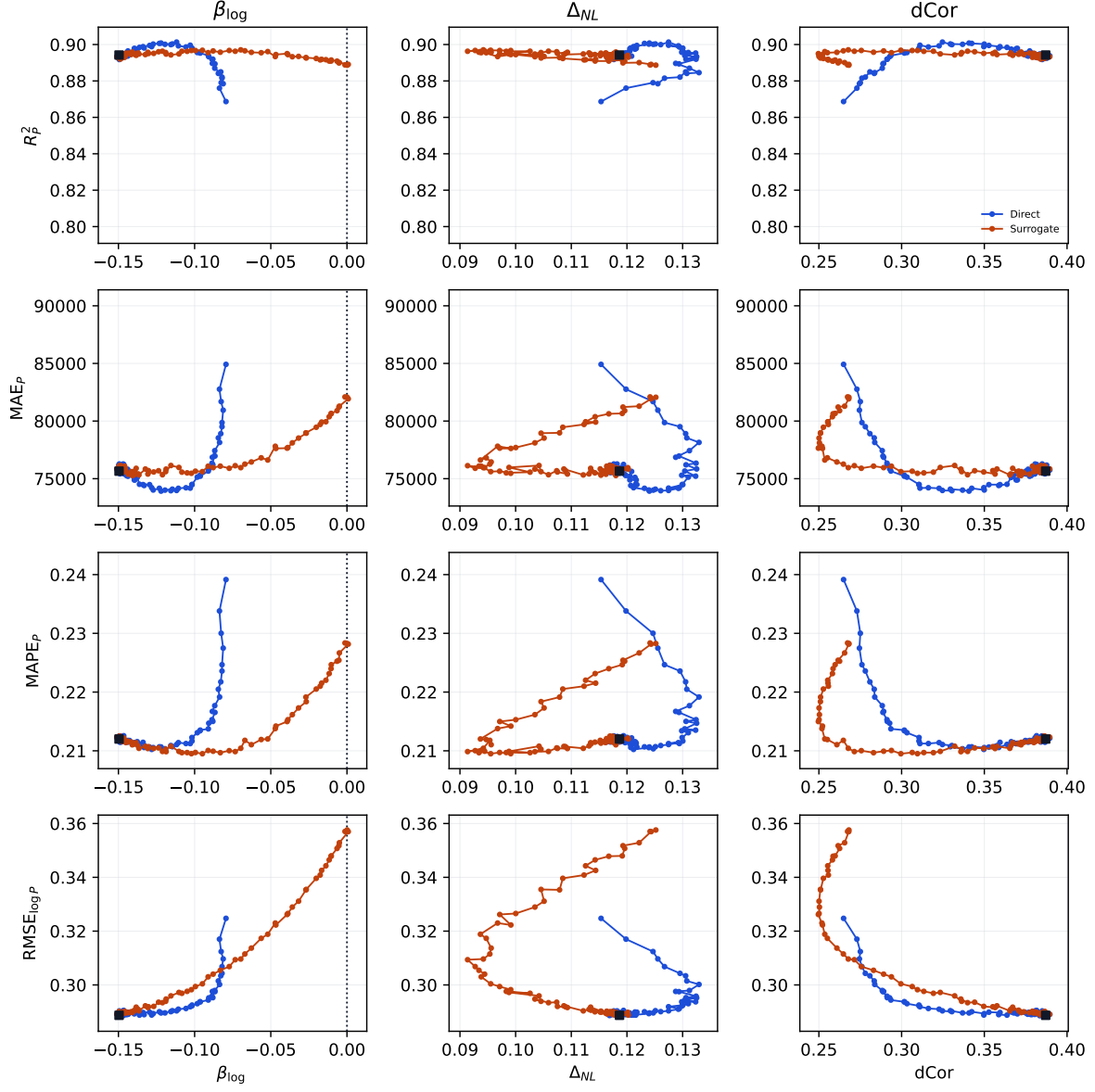


Figure 13: Held-out mechanism/residual diagnostics (horizontal) versus predictive metrics (vertical). The dotted vertical line at $\beta_{\log} = 0$ is a first-order neutrality reference where applicable; Δ_{NL} and $dCor$ are not given assessor-style zero targets.

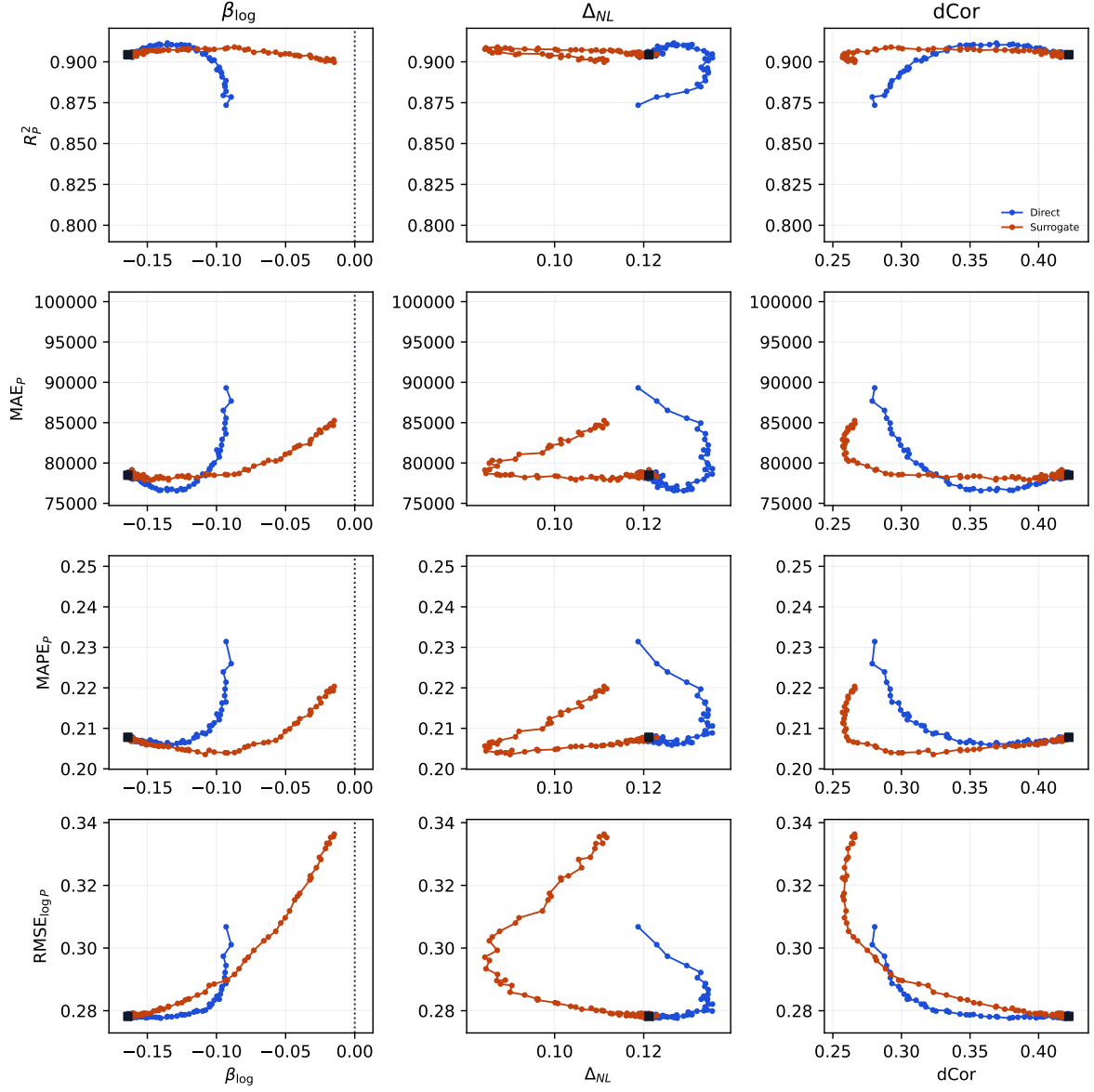


Figure 14: 2025 mechanism/residual diagnostics (horizontal) versus predictive metrics (vertical). The dotted vertical line at $\beta_{\log} = 0$ is a first-order neutrality reference where applicable; Δ_{NL} and $dCor$ are not given assessor-style zero targets.

are exactly unchanged by the choice, and only level-sensitive measures can move at all. Over the blocks on which the two post-hoc paths are traced the largest gap between the two centers is 5.18e-06 — across all three reference cells it is bounded by 8.36e-06 — so at the largest base value the induced log shift is at most 6.266e-06; the largest metric movement the alternative centering produces anywhere is \$1.54 of mean absolute error, and $\beta_{\log}$ moves by 2.78e-17 — machine precision, exactly as the constant-shift argument requires. The sensitivity was therefore evaluated at four base values on each regime for both reference cells instead of as a duplicate full path, and it is reported here with that bound rather than waved away as numerically close.

Development and pooled roots. The rescaling coefficient can be defined on three different samples, and collapsing them would misstate what the comparator does. The fitting-block coefficient is

$$b_T^* = \frac{\widehat{\text{Var}}_T(y)}{\widehat{\text{Cov}}_T(f_0, y)}.$$

It is an in-sample theory diagnostic (Appendix D) and sets no empirical endpoint: for the custom-objective $\rho = 0$ origin it runs from 1.065397 to 1.092817 over the fitting blocks, because in sample that learner’s $\beta_{\log}$ lies between -0.061383 and -0.084934 while out of sample it is about -0.14 to -0.16, so an in-sample calibration would under-correct by roughly a factor of two in $b - 1$. The *primary empirical* root is the development-coordinate one. On the D1 coordinate $\beta_{\log}$ is exactly linear in b , with $\beta_k(b) = b \widehat{\text{Cov}}_{V_k}(f_0, y) / \widehat{\text{Var}}_{V_k}(y) - 1$ per fold and $\beta_{D1}(b)$ their equal-weight mean, so the root is $b_0 = [\frac{1}{7} \sum_k \widehat{\text{Cov}}_{V_k}(f_0, y) / \widehat{\text{Var}}_{V_k}(y)]^{-1}$. Note what that expression is built from: fold-specific *validation*-block covariance and variance, not a training covariance/variance relation. It equals 1.160458 for the custom-objective $\rho = 0$ origin and 1.160367 for the workflow benchmark, and it drives the development coordinate to 1.00e-16 against a target of 10^{-12} . The *third* root is the pooled out-of-fold one at 1.167027 and it is a sensitivity only: the pooled development sample gives the sales in the overlapping validation blocks double observation weight by construction. The grid runs to $b_{\max} = 1.209082$, a 25% overshoot in the adjustment $b - 1$. Every root and bound is computed from development predictions alone. The familiar $1/R^2$ expression, as Appendix D states, is the linear/projection special case rather than the general coefficient for a boosted predictor.

Table 13: Descriptive juxtaposition of the two most-corrected retrained configurations with post-hoc centered rescalings of the *Custom-objective* $\rho = 0$ origin at comparable held-out first-order correction. Held-out evaluation. This comparison is keyed on the held-out coordinate and is descriptive; the comparison matched on the development coordinate is Table 15 and Section 5.5. No configuration is selected or recommended here.

$\beta_{\log}$	Family	Configuration	R_P^2	MAE	Δ_{NL}	dCor
-0.07941	Direct	$\rho = 100$	0.86868	\$84,918	0.11535	0.26493
-0.07905	C-posthoc	$b = 1.080148$	0.89242	\$74,930	0.12946	0.28404
+0.00084	Surrogate	$\rho = 75.4312$	0.88908	\$81,930	0.12434	0.26804
+0.00118	C-posthoc	$b = 1.174235$	0.77962	\$90,429	0.13438	0.25408

Notes. $\beta_{\log}$ is the held-out first-order slope, which is what the two rows in each block are matched on approximately; the post-hoc base value is the tested-grid value whose held-out slope is nearest the retrained configuration’s, not a development-coordinate solve. The development-coordinate solves appear in Table 15 instead. Post-hoc rows are rescalings of a cached prediction array through $f_b(x) = \bar{y}_T + b(f_0(x) - \bar{y}_T)$ with no refitting. Δ_{NL} and dCor carry no assessor-facing reference range and are residual-structure diagnostics, so a lower value is directional only. Values are taken from the retained comparator and reproduction artifacts.

Table 14 reports the comparator of Section 5.4 at the frozen base-value anchors, for the primary base cell on both out-of-time blocks and for the secondary practical base cell on the held-out block. The complete path over all frozen base values, the per-fold summaries with their descriptive chronological spread, the base-value-one bitwise check, the linearity check against the closed form, the root verification, and the anchor-level centering sensitivity are tracked in the repository as part of the frozen post-hoc comparator record. Two properties of the construction are worth restating here because they bound what the table can be read to

show. First, $\beta_{\log}$ is exactly linear in the base value on a fixed evaluation sample, so the post-hoc first-order coordinate is solved rather than searched, and the frozen record verifies the achieved development value against the closed form. Second, the comparator refits nothing: it is an affine transformation of a stored prediction array, which is why it can match a first-order coordinate exactly and cannot alter non-affine conditional-mean structure except incidentally.

Table 14: Centered-spread post-hoc comparator at the frozen base-value anchors. The map is $f_b(x) = \bar{y}_T + b(f_0(x) - \bar{y}_T)$ applied to cached prediction arrays; no model was refit. D1 is the primary development coordinate and D2 the pooled out-of-fold sensitivity, so “D1 root” is the base value that drives the primary development coordinate to first-order neutrality and “D2 root” the same for the sensitivity coordinate. VEI is a May-2026 Exposure Draft diagnostic and is proposed rather than adopted guidance, so no VEI entry is a compliance determination. No base value is selected or recommended.

Anchor	b	R_p^2	MAE	RMSE _{log P}	COD	PRD	MKI	VEI	$\beta_{\log}$	Δ_{NL}	dCor
<i>Panel A: Custom-objective $\rho = 0$ origin rescaled, held-out (primary)</i>											
$b = 1$ (no rescaling)	1.000000	0.8928	\$75,976	0.2894	21.60	1.0690	0.9230	-26.17	-0.1474	0.1164	0.3822
D1 root	1.160458	0.8063	\$87,182	0.3071	22.59	0.9871	1.0726	14.77	-0.0106	0.1343	0.2518
D2 root	1.167027	0.7941	\$88,689	0.3087	22.69	0.9839	1.0787	16.44	-0.0050	0.1344	0.2526
$b_{\max}$	1.209082	0.6926	\$99,978	0.3200	23.50	0.9630	1.1175	26.82	0.0309	0.1338	0.2681
<i>Panel B: Custom-objective $\rho = 0$ origin rescaled, 2025 forward (primary)</i>											
$b = 1$ (no rescaling)	1.000000	0.9039	\$79,139	0.2791	21.23	1.0775	0.9093	-28.41	-0.1612	0.1212	0.4168
D1 root	1.160458	0.7991	\$96,243	0.2939	21.91	0.9934	1.0581	11.37	-0.0266	0.1454	0.2542
D2 root	1.167027	0.7850	\$98,248	0.2955	22.01	0.9900	1.0642	12.97	-0.0211	0.1457	0.2536
$b_{\max}$	1.209082	0.6682	\$112,802	0.3069	22.75	0.9685	1.1029	23.74	0.0142	0.1459	0.2619
<i>Panel C: Ordinary LightGBM (standard raw-label native) rescaled, held-out (secondary, practical)</i>											
$b = 1$ (no rescaling)	1.000000	0.8942	\$75,655	0.2887	21.63	1.0695	0.9230	-26.46	-0.1496	0.1187	0.3871
D1 root	1.160367	0.8024	\$87,658	0.3060	22.61	0.9875	1.0728	14.34	-0.0132	0.1375	0.2531
D2 root	1.167266	0.7892	\$89,277	0.3076	22.73	0.9840	1.0792	16.18	-0.0074	0.1376	0.2537
$b_{\max}$	1.209082	0.6852	\$100,604	0.3188	23.53	0.9633	1.1180	27.33	0.0282	0.1371	0.2678

Notes. Panels A and B rescale the *Custom-objective $\rho = 0$ origin*, the primary post-hoc base because it is the common origin of both penalized families, so rescaling it holds the execution path fixed. Panel C rescales *Ordinary LightGBM (standard raw-label native)* and is the secondary, practical variant; under the frozen convention it determines neither the comparison range nor the choice of comparison targets nor the primary mechanistic test. *Centered-label native L2 (initialization-aligned)* has no post-hoc path. At $b = 1$ the map returns the cached prediction array unchanged, so the first row of each panel is the reference cell itself. VEI is reported in percent and COD in percent; COD lies outside the adopted [5, 15] range throughout. The alternative centering about the fitting block’s mean prediction is a named sensitivity evaluated at these same four base values; its analytic difference from the map used here is constant in x , so it cannot move $\beta_{\log}$, COD, COV or dCor at all. Δ_{NL} and dCor carry no assessor-facing reference range.

F.6 Matched First-Order Targets and Attainment

Frozen target construction and match check. Matching uses the primary development coordinate D1, the equal-weight mean of the seven chronological fold values; the one-sale-one-vote D3 coordinate enters below as the sensitivity. The three families share the development range from -0.138271 to -0.078651 whose upper endpoint is set by Direct, the binding family. Six comparison targets are placed in that range by a deterministic rule: probe equally spaced points, then take as the target the *achieved* development $\beta_{\log}$ of the fitted Direct configuration nearest the probe. Because the target is an achieved value rather than the probe, Direct matches exactly by construction and no selection discretion enters. Table 15 lists the six targets and the configuration each family used; every one of them is an actually fitted model or an exactly solved rescaling, and no targeted new fit was required. The materiality and matching tolerance is $\tau = 0.002$. It was frozen in the matched-configuration record before any held-out or 2025 metric was read. That ordering is a strength of the design rather than a caveat, and it is what makes both the attainment adjudication below and the materiality classification of the D3 sensitivity credible: the worst matching gap over the twenty-four matched cells is 1.0569e-03 and lies inside τ . The $j = 0$ row is an identity check rather than a result — Direct at $\rho = 0$, Surrogate at $\rho = 0$ and the post-hoc map at $b = 1$ are the same object — and the largest relative disagreement anywhere in that check, over all seventeen metrics and four evaluations, is 4.42e-12.

One qualification governs every reading of the outcome columns. The configurations are matched on the *development* coordinate, and nothing in the design equalises their held-out or forward $\beta_{\log}$; matching a development statistic does not transport to an evaluation block. The out-of-time differences below are therefore differences between configurations chosen to approximately match the frozen development first-order coordinate, not effects measured at exactly equal out-of-sample first-order correction.

Distinct-sale sensitivity. Because the fold-6 and fold-7 validation blocks overlap, the primary D1 coordinate carries an observation weighting that was never chosen deliberately, and the frozen design requires the row-balanced D3 coordinate as a sensitivity. It is not a small perturbation. The largest disagreement between the two coordinates is 0.004867 in $\beta_{\log}$ units, larger than τ ; the common support shifts to the range from -0.137744 to -0.076705; and re-running the identical deterministic construction on the 130,165 distinct development sales reselects 13 of the 24 matched configurations — eleven of them exact re-solves of the one-parameter post-hoc map, which must move when the coordinate does, and two different fitted penalty values, both at $j = 2$. Of 315 headline sign comparisons, 259 agree in sign, 41 fall below the materiality floor on both coordinates, 5 are structural zeros forced by the D1 match, and 10 are material sign flips; raw sign agreement across all 315 is 0.9365. Rank order is not preserved between the two coordinates. That reordering should not be read as the overlap scrambling the paths: the $\beta_{\log}$ paths of both retrained families are non-monotone in ρ under *both* coordinates, while both post-hoc families are monotone under both, so a shift of a few thousandths reorders configurations that were already close to tied. The subsection’s high-level conclusions survive the change of coordinate: no Δ_{NL} comparison flips sign under D3 anywhere, and no flip touches the Direct degradation at $j = 5$. What does not survive is the fine ordering of near-ties. Ten pairwise comparisons cross a frozen floor and reverse sign, all of them in small differences — the largest is held-out mean absolute error at $j = 2$, which moves from +\$15 to -\$149 against an error of about \$76,000 — and one of them is the $j = 5$ price-scale tie reported above: on D3 the Surrogate-minus-post-hoc difference is +0.00055 rather than -0.00035, above the floor and pointing the other way. That particular comparison is therefore coordinate-sensitive and is not read as a result. The seven chronological folds are overlapping windows rather than independent replicates, so no statement here is a significance statement, and no fold spread is read as a standard error.

Tables 15 and 16 carry the design and the outcomes behind Section 5.5: the six comparison targets inside the three-way development range with the configuration each family used, the held-out outcome at those targets, and the three extended targets outside the range with an explicit attainment state for every family. The D1 coordinate is primary throughout and the one-sale-one-vote D3 coordinate is the sensitivity; the D3 reselection counts and sign-agreement classification reported in Section 5.5 come from the frozen D3 branch, which re-runs the identical deterministic construction on the row-balanced coordinate and reproduces its targets from the pre-outcome record. The equal-weight and forward-year outcome columns, the pairwise difference tables, the crossing locations, the origin-identity check and the D3 twins of all four tables are tracked in the repository as part of the frozen matched-target record.

F.7 Retransformation Sensitivity

Predictions are returned to the price scale by direct exponentiation, Equation (2). That is an operational retransformation and not a conditional-mean correction, so it is a convention rather than a result. A frozen sensitivity therefore re-evaluates every display configuration under a Duan smearing factor

$$s = \frac{\sum_i w_i \exp(u_i)}{\sum_i w_i}, \quad u_i = \log P_i - f(x_i),$$

applied as $f(x_i) \mapsto f(x_i) + \log s$. The residual convention here is the negative of the one used by the residual-dependence diagnostics, where $e_i = f(x_i) - \log P_i$; the two are not interchangeable. The factor is estimated from development out-of-fold residuals under the one-sale-one-vote weighting and is never re-estimated on an evaluation block.

The consequences separate by construction. Valuation-level ratios—median, mean and weighted-mean—are multiplied exactly by s . COD, COV, PRD, PRB, MKI, VEI, $\beta_{\log}$, Δ_{NL} and $\text{dCor}(e, y)$ are invariant to a common multiplicative factor as a matter of algebra, so no vertical-equity, uniformity, or residual-structure result in this paper depends on the retransformation choice. The frozen sensitivity corroborates

Table 15: Matched first-order comparison on the primary development coordinate D1. Panel A gives the six comparison targets and the configuration each family used; Panel B gives the held-out outcome at those targets for the three primary families. Every configuration is an actually fitted model or an exactly solved rescaling, and no metric is interpolated between penalty values. The match is on the *development* coordinate and does not equalise held-out first-order correction. No configuration is selected or recommended.

<i>Panel A: comparison targets and the matched configurations</i>							
j	Target $\beta_{\log}$ (D1)	Direct ρ	Surrogate ρ	C-posthoc b	A-posthoc b		
0	-0.138271	0	0	1.000000	0.999922		
1	-0.126490	0.7197	0.1326	1.013671	1.013592		
2	-0.114733	1.6768	0.4095	1.027316	1.027235		
3	-0.102033	3.9069	0.8286	1.042053	1.041972		
4	-0.090902	9.1030	1.4563	1.054971	1.054888		
5	-0.078651	86.8511	2.5595	1.069186	1.069103		
<i>Panel B: held-out outcome at those targets</i>							
j	Family	R_P^2	MAE	RMSE $_{\log P}$	COD	Δ_{NL}	dCor
0	Direct	0.8928	\$75,976	0.2894	21.60	0.1164	0.3822
	Surrogate	0.8928	\$75,976	0.2894	21.60	0.1164	0.3822
	C-posthoc	0.8928	\$75,976	0.2894	21.60	0.1164	0.3822
1	Direct	0.8984	\$74,516	0.2888	21.60	0.1216	0.3637
	Surrogate	0.8954	\$75,416	0.2909	21.50	0.1132	0.3660
	C-posthoc	0.8972	\$74,897	0.2893	21.55	0.1192	0.3625
2	Direct	0.9006	\$74,167	0.2900	21.57	0.1231	0.3424
	Surrogate	0.8948	\$75,831	0.2948	21.36	0.1035	0.3358
	C-posthoc	0.9000	\$74,151	0.2895	21.52	0.1218	0.3438
3	Direct	0.8993	\$74,164	0.2910	21.70	0.1292	0.3227
	Surrogate	0.8963	\$75,505	0.2976	21.40	0.0983	0.3134
	C-posthoc	0.9009	\$73,725	0.2900	21.52	0.1243	0.3248
4	Direct	0.8958	\$75,296	0.2937	21.93	0.1298	0.3017
	Surrogate	0.8955	\$76,083	0.3030	21.39	0.0938	0.2938
	C-posthoc	0.8999	\$73,755	0.2908	21.55	0.1263	0.3096
5	Direct	0.8760	\$82,764	0.3170	23.97	0.1198	0.2730
	Surrogate	0.8963	\$76,128	0.3094	21.40	0.0914	0.2714
	C-posthoc	0.8966	\$74,236	0.2920	21.61	0.1282	0.2944

Notes. Targets are placed by a deterministic rule: probe equally spaced points in the three-way development range, then take as the target the achieved development $\beta_{\log}$ of the fitted Direct configuration nearest the probe. The target is therefore an achieved value rather than the probe, Direct matches exactly by construction, and no selection discretion enters. Surrogate rows use the nearest actually fitted penalty value and the post-hoc rows an exact solve, all within the frozen tolerance $\tau = 0.002$. The $j = 0$ row is an origin identity check: Direct at $\rho = 0$, Surrogate at $\rho = 0$ and the post-hoc map at $b = 1$ are the same object. A-posthoc is the rescaling of *Ordinary LightGBM* (*standard raw-label native*) and is carried as a secondary, practical variant; per the frozen convention it determines neither the comparison range, nor the target selection, nor whether a target exists, and its outcome columns are therefore not reported here. Direct and Surrogate penalty magnitudes are not comparable across families. COD is in percent and lies outside the adopted [5, 15] range throughout. Δ_{NL} and dCor are residual-structure diagnostics with no assessor-facing reference range.

Table 16: Extended development targets, which lie outside the three-way common range, with an explicit attainment state for every family. Held-out evaluation. Rows recorded as not attained carry blank outcome columns: they are neither dropped nor filled by interpolation. The states describe the penalty grid this experiment evaluated, not the objectives and not penalty values the experiment never examined.

Target $\beta_{\log}$ (D1)	Family	Configuration	State	Max attained $\beta_{\log}$	R_P^2	MAE	RMSE $_{\log P}$	COD	Δ_{NL}	dCor
-0.06	Direct	—	NOT_ATTAINED	-0.078651	—	—	—	—	—	—
	Surrogate	5.9636	attained	—	0.8934	\$77,813	0.3230	21.81	0.0968	0.2518
	C-posthoc	1.090830	attained	—	0.8868	\$75,847	0.2944	21.75	0.1306	0.2752
-0.03	Direct	—	NOT_ATTAINED	-0.078651	—	—	—	—	—	—
	Surrogate	28.1177	attained	—	0.8917	\$79,939	0.3443	22.84	0.1126	0.2553
	C-posthoc	1.125644	attained	—	0.8569	\$80,444	0.2999	22.10	0.1332	0.2559
0.00	Direct	—	NOT_ATTAINED	-0.078651	—	—	—	—	—	—
	Surrogate	—	NOT_ATTAINED	-0.018609	—	—	—	—	—	—
	C-posthoc	1.160458	attained	—	0.8063	\$87,182	0.3071	22.59	0.1343	0.2518

Notes. Configuration is a fitted penalty value for the Surrogate rows and an exactly solved base value for the post-hoc rows. The “Max attained” column is filled only for rows recorded as not attained, and reports the largest development first-order correction that family reached over the evaluated frozen path. Over that path the Direct family did not attain any of the three extended targets and the Surrogate family did not attain zero; both post-hoc families attain all three exactly, because a one-parameter rescaling can be solved for any first-order target. Reaching a target is not by itself an advantage: driving the post-hoc coordinate to neutrality costs the price-scale accuracy recorded in Table 14. COD is in percent; Δ_{NL} and dCor carry no assessor-facing reference range.

that invariance numerically on the configurations it checks: exhaustively for the level, uniformity, vertical-equity and distance-correlation measures, and on a smaller checked subset for Δ_{NL} , whose guarantee rests on the algebra rather than on that subset. R_P^2 , MAE and MAPE are not invariant and move nontrivially. RMSE $_{\log P}$ is defined on the log scale and is not a price-scale rescaling quantity, so the frozen design does not recompute it under smearing.

Table 17 reports the movement at the three matched first-order configurations of Table 8. The factor is not common across the arms—it is largest for the Surrogate and smallest for the post-hoc comparator—so direct exponentiation does not affect them equally. Under the sensitivity the price-scale ordering of the Surrogate and the post-hoc comparator reverses on both R_P^2 and MAE. The price-scale half of that matched comparison is therefore convention-dependent and is reported as such rather than as a property of the mechanisms; the residual-structure half is not, since the Δ_{NL} and dCor entries of Table 8 are unchanged. The frozen sensitivity was evaluated over the whole frozen configuration set rather than this table alone; what is reported here is the matched-target illustration, because the purpose is to show that the price-scale half of that comparison is convention-dependent, not to introduce a second ranking.

Table 17: Retransformation sensitivity at the matched first-order targets, development coordinate D1 at the strongest common target, held-out evaluation. Baseline columns reproduce Table 8; smeared columns apply the frozen Duan factor s to the same stored predictions. Nothing is refitted, no configuration is selected or recommended, and the table is a convention check rather than a result.

Family	s	R_P^2 base	R_P^2 smeared	MAE base	MAE smeared
Direct	1.0636	0.8760	0.8498	\$82,764	\$87,979
Surrogate	1.0879	0.8963	0.9010	\$76,128	\$73,908
C-posthoc	1.0524	0.8966	0.8860	\$74,236	\$75,548

One asymmetry is recorded rather than resolved. The factor is estimated on development out-of-fold residuals, whereas the 2025 forward block is evaluated from a production refit that has no out-of-fold analogue, so the development-estimated factor is applied there unchanged. That is an explicit assumption of the sensitivity, not a validated property of the forward period.

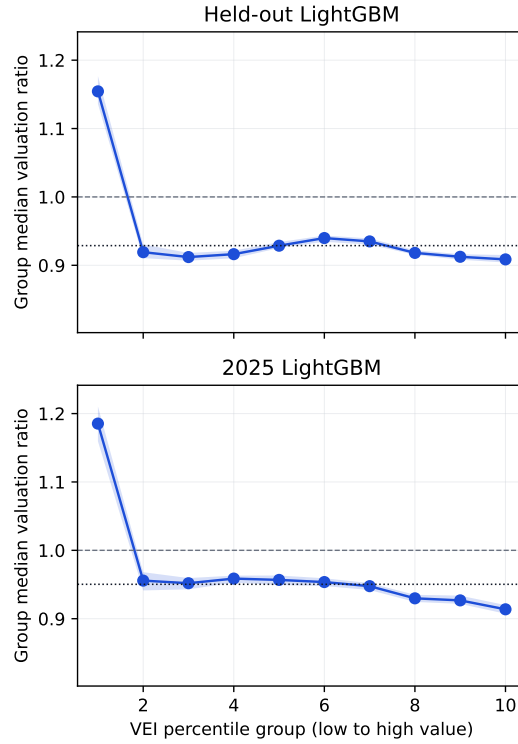


Figure 15: VEI percentile-group median valuation ratios for ordinary LightGBM on the held-out and 2025 samples. The dashed line marks a valuation ratio of 1; the dotted line marks the overall sample median ratio. The shaded ribbons are descriptive, deterministic percentile-bootstrap intervals for the plotted group medians at a nominal 90% level, drawn only to convey sampling variability in this visualization. They are *not* the rank-based order-statistic median interval that the May-2026 IAAO exposure draft specifies for its own VEI procedure in Appendix D.2, and they are *not* a standards-facing bootstrap-sensitivity calculation. No exposure-draft VEI verdict is attached to this figure.

F.8 VEI Percentile-Group Profiles

G Implementation Considerations for Future Use

For a later deployment study, the following sequence would keep model comparisons auditable. The final step is a prospective decision this study does not make.

1. Build a defensible baseline model using verified sales, appropriate time adjustment, relevant property features, and local market information, fitted as the office would ordinarily run it. This is the workflow benchmark against which the rest is described.
2. Reserve future observations for held-out and forward evaluation and create rolling or blocked validation folds that respect sale chronology.
3. Diagnose the first-order mechanism on that benchmark before changing the objective. Report baseline prediction and valuation-level results. Add COD, PRD, PRB, MKI, VEI, value-group ratio profiles, $\beta_{\log}$, Δ_{NL} , and dCor.
4. Fit the Direct and Surrogate covariance-guided objectives over prespecified logarithmic grids of ρ , including $\rho = 0$ as an implementation control and as the within-family origin against which penalty effects are read.
5. Summarize the complete regularization paths across chronological validation folds, including fold-level variability, and report the paths themselves rather than a preferred segment of them.
6. Compare the training-time configurations against the simple post-hoc alternatives on the same predictions — constant level calibration and a validation-fitted centered spread map — using identical temporal splits and diagnostics.
7. Compare families at matched first-order correction rather than at matched penalty, and read predictive accuracy on both the price and log scales alongside the assessor-facing vertical-equity measures and the broader residual-structure diagnostics. State the retransformation convention under which any price-scale comparison is made.
8. Refit the same fixed paths on the full development samples and evaluate them descriptively on the held-out and forward periods without changing the grid or base learner.
9. Repeat the diagnostics by geography, property class, value range, and time before any later deployment-oriented model-selection decision.
10. If deployment eventually requires a single operating point, define that decision prospectively instead of reading it off the paths above: fix the predictive, valuation-level, and equity criteria, their acceptance thresholds, and the tie-breaking rule in advance, then validate that rule out of time on periods not used to formulate it. The retrospective path characterization reported here does not supply such a rule and is not a substitute for one.

A future implementation should save the penalty definition, full grid, validation predictions, held-out and forward predictions, and all ratio-study outputs. This documentation would allow the correction to be re-evaluated in later assessment cycles.

H Reproducibility and Archived Artifacts

This appendix states what the repository archives, what a reader can therefore check, and what cannot be reproduced from public material. A paper-specific guide, `paper/REPRODUCIBILITY.md`, carries the path-level detail, the machine-readable inventory and the exact commands; only the substance is summarized here. Three levels are distinguished, and they are deliberately not collapsed into a single claim of reproducibility.

Table 18: Different pipeline components address different problems.

Component	Primary role
Better property, spatial, and temporal data	Reduce omitted structure and improve the conditional valuation model; may improve accuracy and equity together
Constant level calibration	Move the overall ratio level toward its required target; does not change log-residual covariance
Covariance-guided regularization	Discourage a global first-order log-ratio/log-price trend during model training
Residual-structure and local diagnostics	Use $\beta_{\log}$, Δ_{NL} , distance correlation, and ratio-shape diagnostics to distinguish first-order, non-affine conditional-mean, and broader dependence patterns; use stratified diagnostics to detect geographic or property-class patterns hidden by countywide summaries.
Temporal validation and uncertainty analysis	Characterize how path behavior changes across market periods; by itself, chronological validation does not establish a temporally invariant penalty range

Auditability of the reported results. Every number printed in this paper can be re-derived without refitting anything. The manuscript source, the figures it includes, the frozen specifications and reference conventions, the provenance records, and a numeric ledger are tracked in the repository. The ledger binds each printed value to a tracked artifact by content hash, together with the selector that locates it, the transform applied to it and the rounding rule used to print it; a validator reopens each artifact, verifies its hash against both the file and the frozen manifest, re-executes the selector and compares the result with what the source prints. The executed metric code, including the ratio-study measures, the nonlinearity-gap estimator and the distance-correlation call site, is tracked and is byte-identical at every provenance commit.

Reconstruction from frozen derived artifacts. The derived evidence behind the reported comparisons is tracked and can be re-examined without the raw extract. This covers the matched first-order design and its outcomes, with the extended targets, pairwise differences, crossing locations, origin-identity check and the one-sale-one-vote sensitivity twins; the post-hoc comparator path over all frozen base values, with its per-fold summaries, closed-form linearity check, root verification and centering sensitivity; the zero-penalty control and the parity ladders; the retransformation-sensitivity, PRB and Exposure-Draft VEI inference tables; and the distance-correlation implementation record. What this level does not include is the per-configuration fold, held-out and forward prediction arrays, which are record-level outputs derived from restricted assessor data and are not released.

End-to-end rerun. A full rerun of the empirical pipeline is not publicly reproducible. It requires the Cook County extract identified in Section 4.1, which is hash-pinned but not redistributed and for which no public availability is claimed, together with the executed run roots holding the stored predictions and the combined path table. The frozen evidence manifest records each of those inputs by path, byte length and content hash, so the reader can confirm which file was used; it does not put the file itself in the reader’s hands. A reader without authorized access to that exact input can audit and reconstruct the reported results, but cannot refit the models.

Two limitations belong with this account. The recorded environment is the one the executed pipeline observed and not a lockfile: the interpreter and the seven libraries that matter for the results are pinned to exact versions in two independent frozen records, while the repository requirement file states only lower bounds and does not reconstruct that environment. And the source state that generated the frozen artifacts is bounded rather than recovered, for the reason given in Appendix E: the generating trees were dirty and the uncommitted portion was never archived.

Nothing in this appendix is a statement about operational readiness. Before operational use, the same analysis should report block-bootstrap intervals and ratio curves by township, property class, value range, and time, so that model-selection uncertainty, implementation effects and local variation are separated. This paper reports none of those.

Data, Code, and Institutional Statement

The primary application uses Cook County Assessor’s Office data together with a research translation of the office’s published residential modeling workflow. The analysis code, the frozen specifications and conventions, the derived result tables behind every reported comparison, the provenance records and the numeric ledger are tracked in the public repository, and `paper/REPRODUCIBILITY.md` is the entry point for auditing this paper specifically; the repository’s general development README describes a broader legacy workflow and is not the protocol behind these results. The raw CCAO extract is not redistributed. It is identified by content hash and byte length rather than by a version string, and no public availability is claimed for it, so an external reader can verify which input was used but cannot obtain it here. Consequently the reported results can be audited and reconstructed from tracked artifacts, while refitting the models requires authorized access to that exact input. The per-record prediction artifacts are not released. The results reported here are research findings from a research translation of the CCAO workflow, not official CCAO assessment decisions, and the paper makes no claim that CCAO has adopted the correction or placed it into production.